INCH-POUND
MIL-M-38510/656A
12 October 2005

SUPERSEDING
MIL-M-38510/656
31 March 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, HIGH-SPEED CMOS, FLIP-FLOPS, MONOLITHIC SILICON

Reactivated after 12 Oct. 2005 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF 38535.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, high speed CMOS, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN). For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535 (see 6.3).

1.2 Part or identifying number (PIN). The PIN is in accordance with MIL-PRF-38535 and as specified herein.

1.2.1 Device types. The device types are as follows:

Device type	Circuit
01	Octal D-type flip-flop with clear
02	Octal three-state D-type flip-flop
03	Octal D-type flip-flop with enable
04	Octal three-state D-type flip-flop
05	Octal three-state D-type flip-flop with inverting outputs
06	Octal three-state D-type flip-flop with inverting outputs
52	Octal three-state D-type flip-flop with TTL compatible inputs

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

Outline letter	Descriptive designator	Terminals	Package style
R	GDIP1-T20 or CDIP2-T20	20	Dual-in-line
S	GDFP2-F20 or CDFP3-F20	20	Flat pack
2	CQCC1-N20	20	Square leadless chip carrier
X	See figure 4	20	Flat pack

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1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 V dc to +7.0 V dc
DC input voltage range (V_{IN})	-0.5 V dc to V_{CC} + 0.5 V dc
DC output voltage range (V_{OUT})	-0.5 V dc to V_{CC} + 0.5 V dc
Clamp diode current (I_{IK}, I_{OK})	±20 mA
DC output current per pin (I_{OUT})	±25 mA
DC V_{CC} or GND current per pin (I_{CC}, I_{GND})	±50 mA
Storage temperature range (T_{STG})	-65° to +150°C
Maximum power dissipation, (P_D):	
Device types 01 and 03	300 mW
Device types 02, 04, 05, 06, and 52	420 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC})	See MIL-STD-1835
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Device types 01, 02, 03, 04, 05, 06:

Maximum input low voltage (V_{IL})	0.3 V at V_{CC} = 2.0 V 0.9 V at V_{CC} = 4.5 V 1.2 V at V_{CC} = 6.0 V
Minimum input high voltage (V_{IH})	1.5 V at V_{CC} = 2.0 V 3.15 V at V_{CC} = 4.5 V 4.2 V at V_{CC} = 6.0 V
Supply voltage range (V_{CC})	2.0 V dc to 6.0 V dc
Output voltage range (V_{OUT})	0.0 V dc to V_{CC}
Operating temperature range (T_C)	-55° to +125°C
Maximum input rise and fall times (t_r, t_f):	
V_{CC} = 2.0 V	1000 ns
V_{CC} = 4.5 V	500 ns
V_{CC} = 6.0 V	400 ns

Device type 52:

Maximum input low voltage (V_{IL})	0.8 V at V_{CC} = 4.5 V to 5.5 V
Minimum input high voltage (V_{IH})	2.0 V at V_{CC} = 4.5 V to 5.5 V
Supply voltage range (V_{CC})	4.5 V dc to 5.5 V dc
Output voltage range (V_{OUT})	0.0 V to 5.5 V
Operating temperature range (T_C)	-55° to +125°C
Maximum input rise and fall times (t_r, t_f):	
V_{CC} = 4.5 V	500 ns

Setup time before clock (t_{SU}):	
All devices	20 ns minimum
Hold time (t_h):	
Device types 01, 03, 04, and 06	5 ns minimum
Device types 02, 05, and 52	10 ns minimum
Clock pulse (t_{CLK}):	
Device types 02, 04, 05, and 06	16 ns minimum
Device type 01	18 ns minimum
Device types 03 and 52	20 ns minimum
Removal time (t_{REM}) (device type 01 only)	20 ns minimum

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard Microcircuits.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.3.2 Truth tables. The truth tables shall be as specified on figure 2.

3.3.3 Switching time test circuit and waveforms. The switching time test circuit and waveforms shall be as specified on figure 3.

3.3.4 Schematic circuits. The schematic circuits shall be maintained by the manufacturer and made available to the qualifying activity or preparing activity upon request.

3.3.5 Case outlines. The case outlines shall be as specified in 1.2.3, figure 4, and MIL-STD-1835.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range.

3.6 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 38 (see MIL-PRF-38535, appendix A).

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4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535 and shall be conducted on all devices prior to qualification and conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1015 of MIL-STD-883.

b. Delete the sequence specified as interim (pre-burn-in) electrical parameters through interim (post-burn-in) electrical parameters of table IA of MIL-PRF-38535 and substitute lines 1 through 7 of table II herein.

c. Burn-in (method 1015 of MIL-STD-883).

(1) Unless otherwise specified in the manufacturers QM plan for static tests (test condition A), ambient temperature (T_A) shall be +125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

i. For static burn-in I, all inputs shall be connected to GND. Outputs may be open or connected to $V_{CC}/2$. Resistors R1 are optional on both inputs and open outputs, required on outputs connected to $V_{CC}/2$. Resistors R1 = 680Ω to 47 kΩ.

ii. For static burn-in II, all inputs shall be connected through the R1 resistors to V_{CC}. Outputs may be open or connected to $V_{CC}/2$. Resistors are optional on open outputs, required on outputs connected to $V_{CC}/2$. Resistors R1 = 680Ω to 47 kΩ.

iii. V_{CC} = 6.0 V ±0.5 V; $V_{CC}/2 = V_{CC}/2$ ±0.5 V.

(2) Unless otherwise specified in the manufacturers QM plan for dynamic test (test condition D), ambient temperature shall be +125°C minimum. Test duration shall be in accordance with table I of method 1015.

i. For dynamic burn-in, all inputs shall be connected through the resistors in parallel to a common clock pulse. Outputs shall be connected to $V_{CC}/2$ ±0.5 V through the resistors. For device types 01 and 03 resistors = 1 kΩ ±5% for outputs. For device types 02, 04-06, and 52 resistors = 680Ω ±5% for outputs. For all devices input resistors = 1 kΩ to 47 kΩ ±20% for inputs.

ii. CP1 = 100 kHz, CP2 = 50 kHz ±50% square wave; duty cycle = 50% ±15%; V_{IH} = 4.5 V to V_{CC}; V_{IL} = 0.0 V ±0.5 V; t_{THL}, t_{TLH} < 500 ns.

iii. V_{CC} = 6.0 V ±0.5 V; $V_{CC}/2 = V_{CC}/2$ ±0.5 V.

d. Interim and final electrical test parameters shall be as specified in table II.

e. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer’s option, be performed separately or included in the final electrical parameter requirements.

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4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failure shall be cumulative for determining the PDA.

c. The PDA for class B devices shall be in accordance with MIL-PRF-38535 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics, after burn-in, exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in the lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 shall be omitted.

c. Subgroup 4 (C_{IN} and C_{OUT} measurements) shall be measured only for initial qualification and after process or design changes that may affect input capacitance. Capacitance shall be measured between the designated terminal and GND at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End-point electrical parameters shall be as specified in table II herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
High-level output voltage	V_{OH1} 2/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OH} = -20 µA	01-06	2.0V	1.90		V
	V_{OH2} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -20 µA	01-06	4.5V	4.40		V
	V_{OH3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -20 µA	01-06	6.0V	5.90		V
	V_{OH4} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OH} = -4.0 mA	01-06	4.5V	3.7		V
	V_{OH5}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OH} = -5.2 mA	01-06	6.0V	5.2		V
	V_{OH6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -20 mA	52	4.5V	4.4		V
	V_{OH7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OH} = -6.0 mA	52	5.5V	3.7		V
Low-level output voltage	V_{OL1} 2/	V_{IH} = 1.5 V V_{IL} = 0.3 V I_{OL} = 20 µA	01-06	2.0V		0.10	V
	V_{OL2} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OL} = 20 µA	01-06	4.5V		0.10	V
	V_{OL3}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OL} = 20 µA	01-06	6.0V		0.10	V
	V_{OL4} 2/	V_{IH} = 3.15 V V_{IL} = 0.9 V I_{OL} = 4.0 mA	01-06	4.5V		0.4	V
	V_{OL5}	V_{IH} = 4.2 V V_{IL} = 1.2 V I_{OL} = 5.2 mA	01-06	6.0V		0.4	V
	V_{OL6}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 20 mA	52	4.5V		0.1	V
	V_{OL7}	V_{IH} = 2.0 V V_{IL} = 0.8 V I_{OL} = 6.0 mA	52	5.5V		0.4	V
Positive input clamp voltage	V_{IC}+	I_{IN} = 1 mA T_C = +25°C	01-06 52	GND		1.5	V
Negative input clamp voltage	V_{IC}-	I_{IN} = -1 mA T_C = +25°C	01-06 52	OPEN		-1.5	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Input current low	I_{IL}	V_{IN} = GND	01-06	6.0V		-0.1	µA
			52	5.5V		-0.1	µA
Input current high	I_{IH}	$V_{IN} = V_{CC}$	01-06	6.0V		0.1	µA
			52	5.5V		0.1	µA
Short circuit output current	I_{OS1} 2/	V_O = GND V_I = GND	01,03	2.0V	-2	-50	mA
			02,04 05,06		-2	-60	
	I_{OS2} 2/		01,03	4.5V	-15	-150	
			02,04 05,06		-15	-165	
	I_{OS3} 2/		01,03	6.0V	-25	-180	
			02,04 05,06		-25	-210	
	I_{OS4}		01,03	4.0V	-10	-120	
			02,04 05,06,52		-10	-135	
	I_{OS5} 2/	V_I = 5.5 V	52	5.5V	-25	-210	
Supply current quiescent	I_{CC}	V_I = 6.0 V	01-06	6.0V		8	µA
		V_I = 5.5 V	52	5.5V		8	µA
	I_{CCZ}	V_I = 6.0 V	02,04 05,06	6.0V		4.0	µA
		V_I = 5.5 V	52	5.5V		4.0	µA
Three-state output leakage current low	I_{OZL}	V_{OUT} = GND OC = V_{IH} 3/	02,04,05,06	6.0V		-2.0	µA
			52	5.5V		-2.0	
Three-state output leakage current high	I_{OZH}	$V_{OUT} = V_{CC}$ OC = V_{IH} 3/	02,04,05,06	6.0V		+2.0	µA
			52	5.5V		+2.0	
Input capacitance	C_{IN}	T_C = 25°C	01-06 52 4/			10	pF
Clock, control capacitance	C_C	T_C = 25°C	01-06 52			15	pF
Three-state output capacitance	C_O	T_C = 25°C	02,04-06,52			20	pF
Propagation delay times, clear to outputs	t_{PHL1} 5/, 6/	C_L = 50 pF ± 10%	01	4.5V		41	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics – Continued.

Test	Symbol	Conditions 1/ $-55°C \le T_C \le +125°C$ unless otherwise specified	Device type	V_{CC}	Limits		Unit
					Min	Max	
Propagation delay time, clock to outputs	t_{PHL2}, t_{PLH2} 5/, 6/	C_L = 50 pF ± 10%	01	4.5V		37	ns
			02, 52			42	ns
			03			37	ns
			04			42	ns
			05			42	ns
			06			41	ns
Output enable time	t_{PZH}, t_{PZL} 5/, 6/	C_L = 50 pF ± 10% R_L = 1 kΩ ± 10%	02	4.5V		35	ns
			52			41	ns
			04			35	ns
			05			41	ns
			06			35	ns
Output disable time	t_{PHZ}, t_{PLZ} 5/, 6/	C_L = 50 pF ± 10% R_L = 1 kΩ ± 10%	02	4.5V		35	ns
			52			41	ns
			04			35	ns
			05			35	ns
			06			35	ns
Transition time	t_{THL}, t_{TLH} 5/, 6/	C_L = 50 pF ± 10%	01	4.5V		20	ns
			02, 52			16	ns
			03			20	ns
			04			16	ns
			05			16	ns
			06			16	ns
Maximum frequency test	f_{MAX} 5/, 6/	C_L = 50 pF ± 10%, 50% duty cycle	01	4.5V		23	MHz
			02			25	MHz
			52			21	MHz
			03			21	MHz
			04			25	MHz
			05			25	MHz
			06			25	MHz

1/ Complete terminal conditions shall be as specified in table III.

2/ Guaranteed but not tested.

3/ I_{OZL} set internal D flip-flops to high state.
I_{OZH} set internal D flips-flops to low state.

4/ Power dissipation capacitance (C_{PD}) per gate typically equals 25 pF.

5/ Tested at V_{CC} = 4.5 V at +125°C for sample testing and V_{CC} = 4.5 V at +25°C for screening. Guaranteed at other V_{CC} voltages and temperatures, see table IA and IB as appropriate and the exception in 4.4.1d.

6/ See the formula for determining switching times and maximum frequencies shown in tables IA and IB respectively.

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TABLE IA. Calculated dynamic figures at –55°C/+25°C ambient temperature.

V_{CC}	T_A = (°C)	
	125	-55/25
2.0V	5.00X	5.00Y
4.5V	X = 1	0.75X = Y
6.0V	0.85X	0.85Y

Normalized numbers
(+125°C equals 1)

NOTE: The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according to 5/4.

TABLE IB. Calculated f_{MAX} figures –55°C/+25°C ambient temperature.

V_{CC}	T_A = (°C)	
	125	-55/25
2.0V	0.2X	0.2Y
4.5V	X = 1	1.33X = Y
6.0V	1.18X	1.18Y

Normalized numbers
(+125°C equals 1)

NOTE: The 2.0 V and 6.0 V numbers are derived from their 4.5 V integer value. Rounding off according to 5/4.

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TABLE II. Burn-in and electrical test requirements.

Line no.	MIL-PRF-38535 test requirements	Class S device 1/			Class B device 1/		
		Ref. par.	Table III Subgroups 2/	Table IV delta limits 3/	Ref. par.	Table III subgroups 2/	Table IV delta 3/
1	Interim electrical parameters		1			1	
2	Static burn-in I (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
3	Same as line 1		1*	Δ			
4	Static burn-in II (method 1015)	4.2c 4.5.2	Req'd		4.2c 4.5.2	4/ req'd	
5	Same as line 1	4.2e	1*	Δ	4.2e	1*	Δ
6	Dynamic burn-in (method 1015)	4.2c 4.5.2	Req'd			Not req'd	
7	Same as line 1	4.2e	1	Δ			
8	Final electrical parameters		* 1, 2, 3, 9			1*, 2, 3, 9 4/	
9	Group A test requirements	4.4.1	1, 2, 3, 4, 9, 10, 11		4.4.1	1, 2, 3, 4, 9, 10, 11	
10	Group B test when using method 5005 QCI option	4.4.2	1, 2, 3, 9, 10, 11	Δ			
11	Group C end-point electrical parameters	4.4.2	1, 2, 3, 9, 10, 11	Δ	4.4.3	1, 2	Δ
12	Group D end-point electrical parameters	4.4.4	1, 2, 3		4.4.4	1, 2	

1/ Blank spaces indicate tests are not applicable.

2/ * indicates PDA applies to subgroup 1 (see 4.2.1).

3/ Δ indicates delta limits shall be required only on table III subgroup 1, where specified, and the delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either complete subgroup 1 electrical parameter measurements, including delta measurements, within 96 hours after burn-in completion (removal of bias); or may complete subgroup 1 electrical measurements without delta measurements within 24 hours after burn-in completion (removal of bias).

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Device type	01
Case outlines	R, S, and 2
Terminal number	Terminal symbol
1	$\overline{\text{CLEAR}}$
2	Q1
3	D1
4	D2
5	Q2
6	Q3
7	D3
8	D4
9	Q4
10	GND
11	CLOCK
12	Q5
13	D5
14	D6
15	Q6
16	Q7
17	D7
18	D8
19	Q8
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

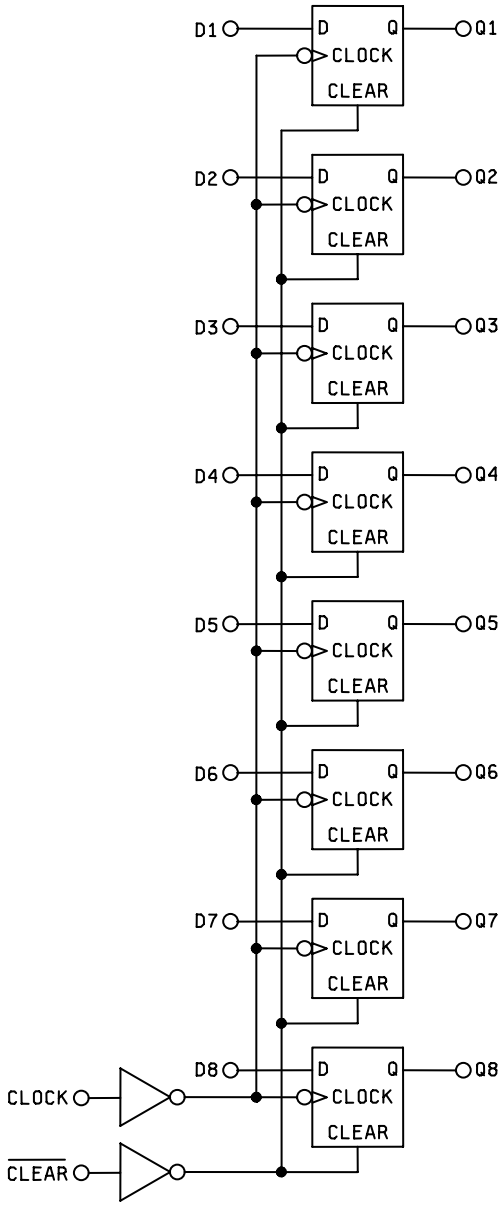


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device types	02 and 52
Case outlines	R, S, and 2
Terminal number	Terminal symbol
1	$\overline{OE}$
2	Q1
3	D1
4	D2
5	Q2
6	Q3
7	D3
8	D4
9	Q4
10	GND
11	CLK
12	Q5
13	D5
14	D6
15	Q6
16	Q7
17	D7
18	D8
19	Q8
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device types 02 and 52

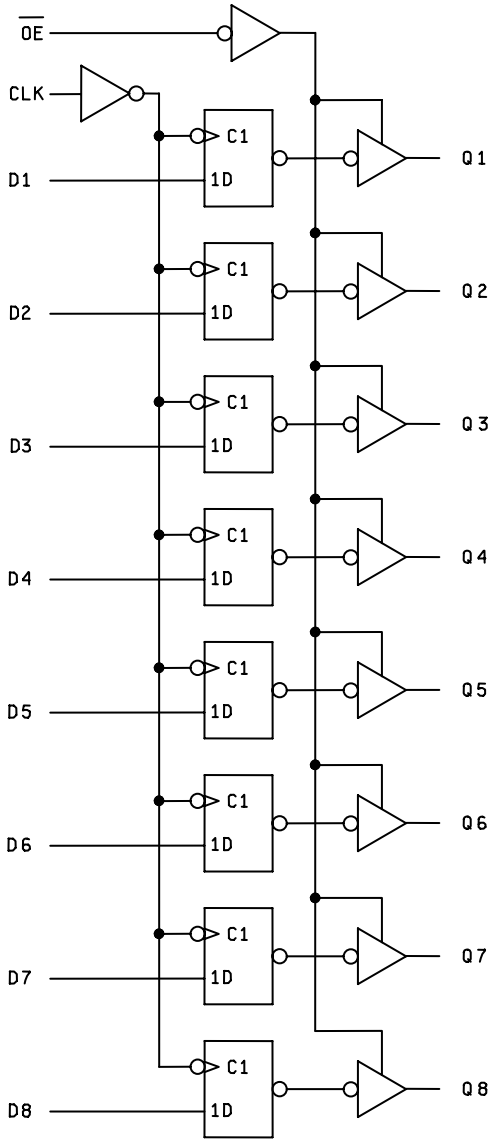


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type	03
Case outlines	R, S, and 2
Terminal number	Terminal symbol
1	$\overline{CE}$
2	Q1
3	D1
4	D2
5	Q2
6	Q3
7	D3
8	D4
9	Q4
10	GND
11	CLK
12	Q5
13	D5
14	D6
15	Q6
16	Q7
17	D7
18	D8
19	Q8
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 03

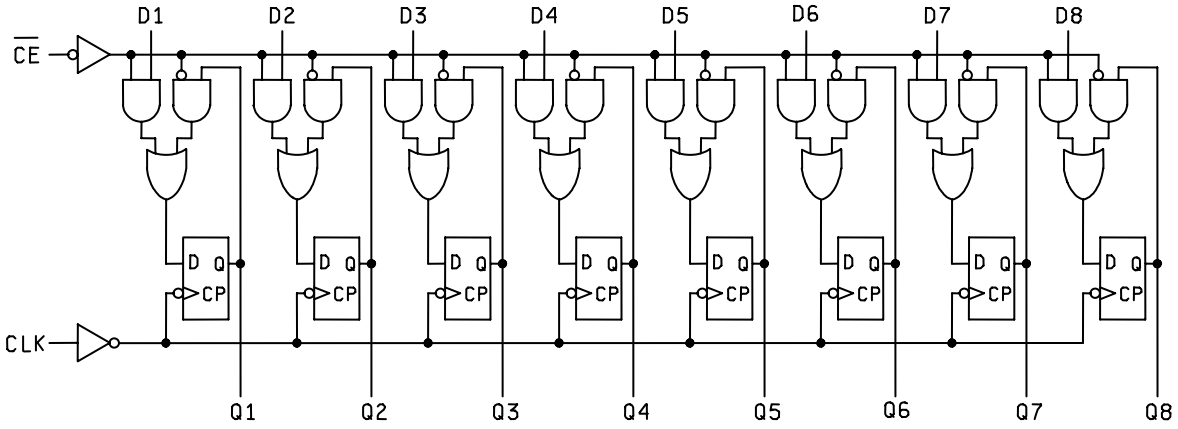


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type	04
Case outlines	R, S, X, and 2
Terminal number	Terminal symbol
1	$\overline{OE}$
2	D1
3	D2
4	D3
5	D4
6	D5
7	D6
8	D7
9	D8
10	GND
11	CLK
12	Q8
13	Q7
14	Q6
15	Q5
16	Q4
17	Q3
18	Q2
19	Q1
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 04

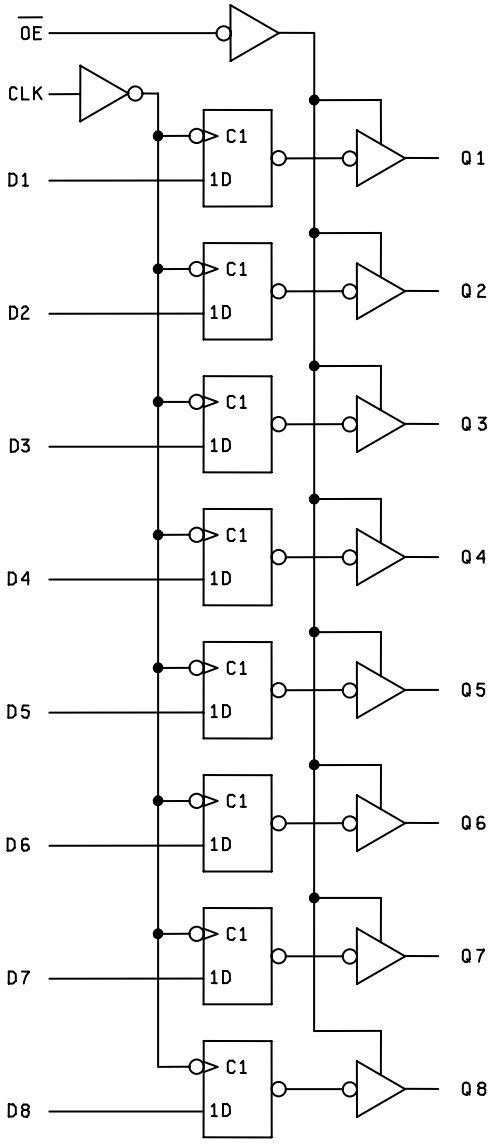


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type	05
Case outlines	R, S, and 2
Terminal number	Terminal symbol
1	$\overline{OE}$
2	$\overline{Q1}$
3	D1
4	D2
5	$\overline{Q2}$
6	$\overline{Q3}$
7	D3
8	D4
9	$\overline{Q4}$
10	GND
11	CLK
12	$\overline{Q5}$
13	D5
14	D6
15	$\overline{Q6}$
16	$\overline{Q7}$
17	D7
18	D8
19	$\overline{Q8}$
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 05

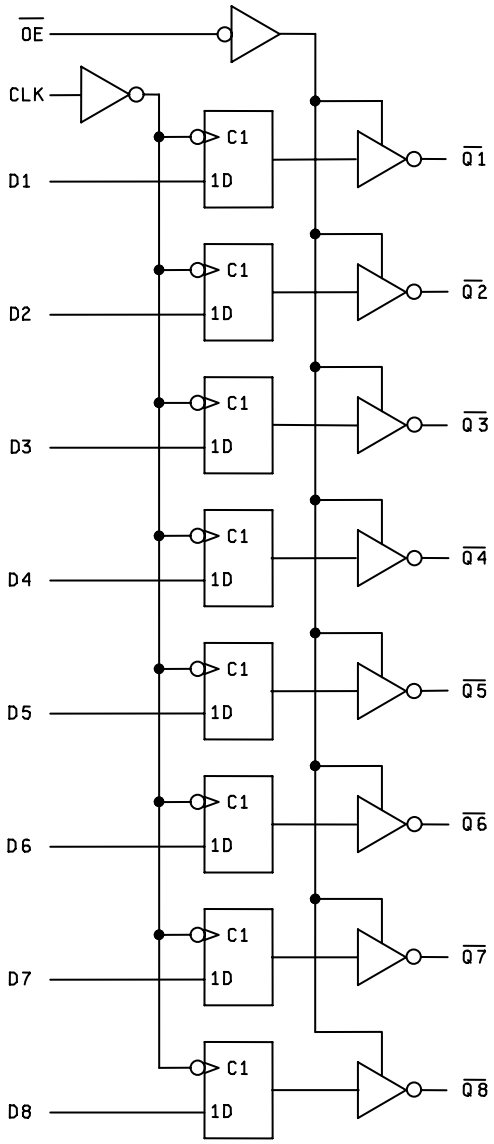


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type	06
Case outlines	R, S, and 2
Terminal number	Terminal symbol
1	$\overline{OE}$
2	D1
3	D2
4	D3
5	D4
6	D5
7	D6
8	D7
9	D8
10	GND
11	CLK
12	$\overline{Q8}$
13	$\overline{Q7}$
14	$\overline{Q6}$
15	$\overline{Q5}$
16	$\overline{Q4}$
17	$\overline{Q3}$
18	$\overline{Q2}$
19	$\overline{Q1}$
20	V_{CC}

FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 06

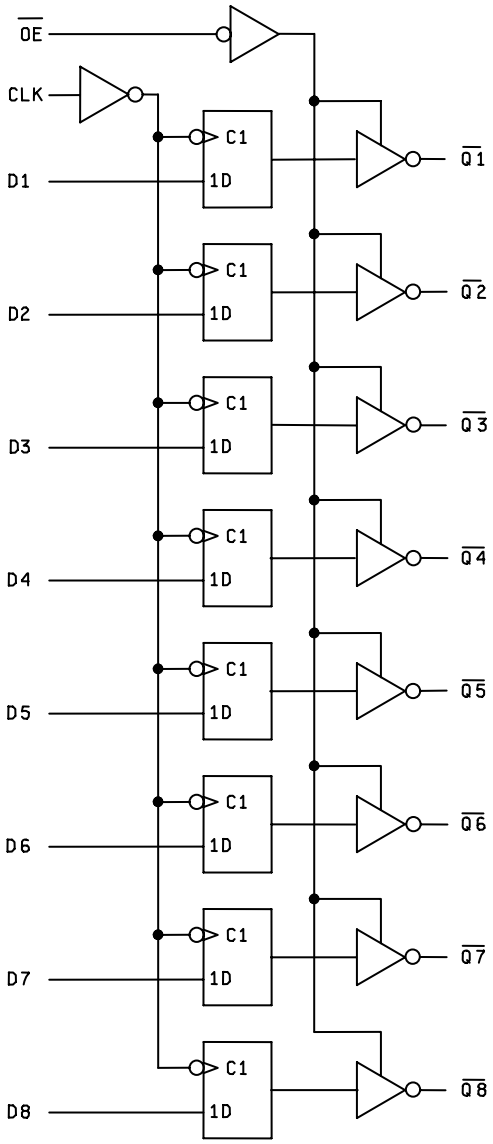


FIGURE 1. Logic diagrams and terminal connections – Continued.

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Device type 01

Inputs			Outputs
Clear	Clock	D	Q
L	X	X	L
H	↑	H	H
H	↑	L	L
H	L	X	Q0

Device types 02, 04, and 52

Inputs			Outputs
Output control	Clock	Data	Q
L	↑	H	H
L	↑	L	L
L	L	X	Q0
H	X	X	Z

Device type 03

Inputs			Outputs
$\overline{CE}$	Clock	Data	Q
H	X	X	Q0
L	↑	H	H
L	↑	L	L
X	L	X	Q0

Device types 05 and 06

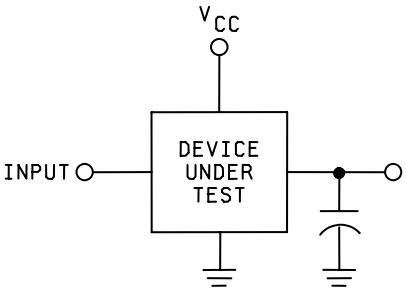
Inputs			Outputs
Output control	Clock	Data	$\overline{Q}$
L	↑	H	L
L	↑	L	H
L	L	X	$\overline{Q0}$
H	X	X	Z

H = High level voltage
L = Low level voltage
X = Don't care
↑ = Low-to-high level transition of the clock
Z = High impedance state
Q0, $\overline{Q0}$ = The level of the output before steady state input conditions were established

FIGURE 2. Truth tables.

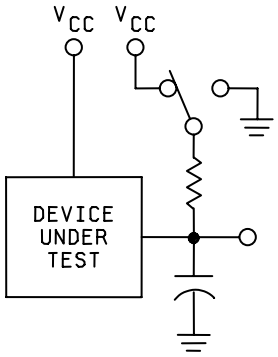
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DEVICE TYPES 01 AND 03



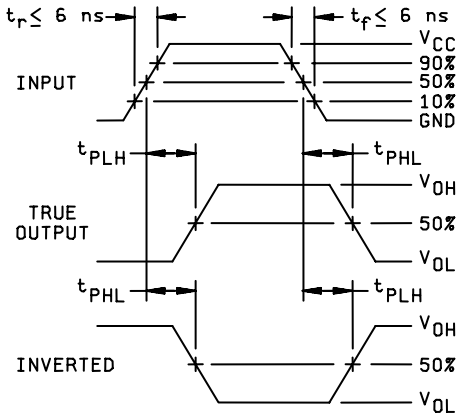
TEST CIRCUIT FOR PUSH PULL OUTPUTS

DEVICE TYPES 02,04,05,06 AND 52

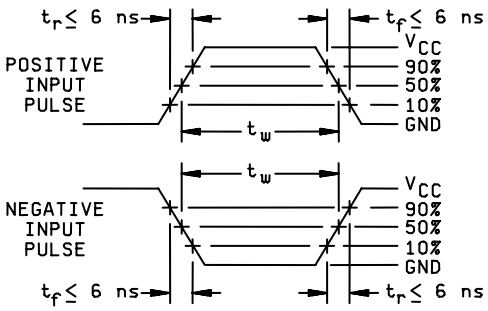


TEST CIRCUIT FOR THREE-STATE
TEST - SEE NOTES 1 AND 2

DEVICE TYPES 01,02,03,04,05,06 AND 52



PROPAGATION DELAY WAVEFORMS

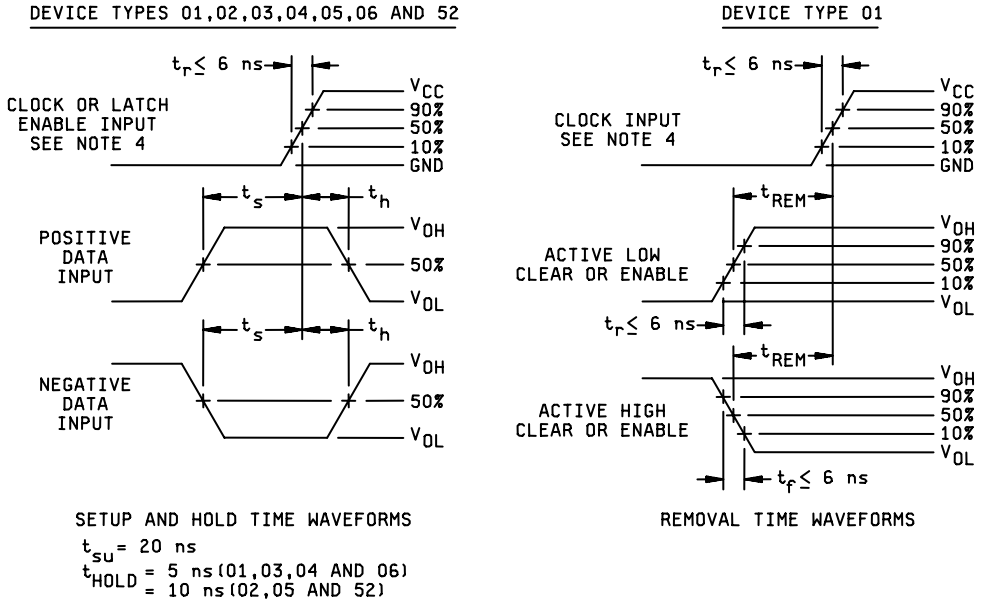


CLOCK INPUT PULSE WIDTH WAVEFORM

MINIMUM PULSE WIDTH = 16 ns
(01 = 18 ns)(03,53 = 18 ns)

FIGURE 3. Switching time test circuit and waveforms.

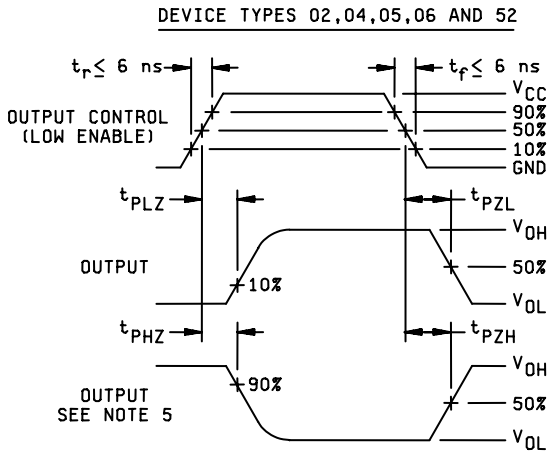
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:57Z
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SETUP AND HOLD TIME WAVEFORMS

t_{su} = 20 ns
t_{HOLD} = 5 ns (01,03,04 AND 06)
= 10 ns (02,05 AND 52)

REMOVAL TIME WAVEFORMS



THREE-STATE OUTPUT ENABLE AND DISABLE WAVEFORMS

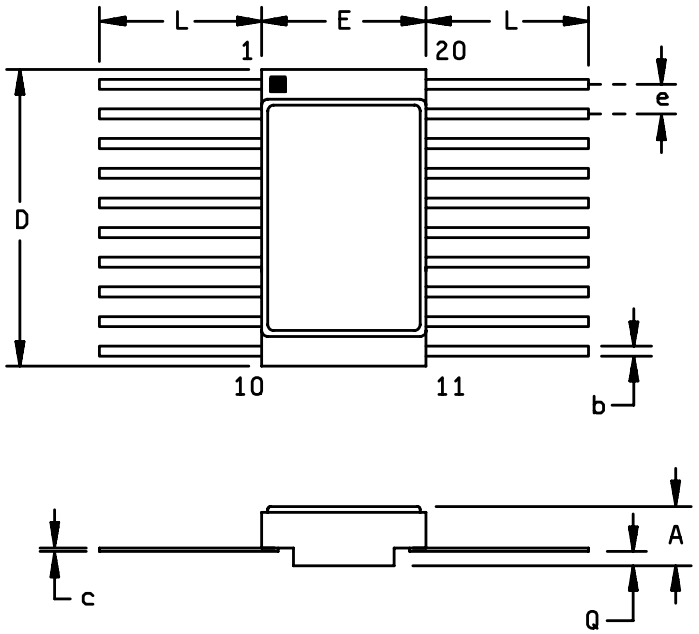
NOTES:

1. C_L includes load and test jig capacitance. See data sheets for values.
2. S1 = V_{CC} for t_{PZL} and t_{PLZ} measurements. S1 = GND for t_{PZH} and t_{PHZ} measurements.
3. Waveforms for negative edge sensitive circuits will be inverted, t_{REM} (20 ns).
4. This waveform is applicable to both three-state and open drain switching time measurements.

FIGURE 3. Switching time test circuit and waveforms – Continued.

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Case X



Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.045	.085	1.14	2.16
b	.015	.019	0.38	0.48
c	.003	.006	0.076	0.152
D	.505	.515	12.83	13.08
E	.275	.285	6.99	7.24
e	.045	.055	1.14	1.40
L	.250	.370	6.35	9.39
Q	.010		0.25	
N	20		20	

FIGURE 4. Case outline.

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TABLE III. Group A inspection for device type 01.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{CLR}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA		1mA	1mA			1mA	1mA		1/	1mA		1mA	1mA			1mA	1mA		GND	$\overline{CLR}$	2/	1.5					V
		2										"										"	D1		"					"
		3										"										"	D2		"					"
		4										"										"	D3		"					"
		5										"										"	D4		"					"
		6										"										"	CLK		"					"
		7										"										"	D5		"					"
		8										"										"	D6		"					"
		9										"										"	D7		"					"
		10										"										"	D8		"					"
V_{IC}(neg)		11	-1mA		-1mA	-1mA			-1mA	-1mA		GND	-1mA		-1mA	-1mA			-1mA	-1mA		1/	$\overline{CLR}$		-1.5					"
		12										"										"	D1		"					"
		13										"										"	D2		"					"
		14										"										"	D3		"					"
		15										"										"	D4		"					"
		16										"										"	CLK		"					"
		17										"										"	D5		"					"
		18										"										"	D6		"					"
		19										"										"	D7		"					"
		20										"										"	D8		"					"
I_{CCH}	3005	21	6V		6V	6V			6V	6V		"	3/		6V	6V			6V	6V		6V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		0.2		8.0			µA
V_{OH3}	3006	23	4.2V	-20µA	4.2V							"	"									"	Q1	5.95		5.95		5.95		V
		24	"			4.2V	-20µA					"	"									"	Q2	"		"		"		"
		25	"					-20µA	4.2V			"	"									"	Q3	"		"		"		"
		26	"							4.2V	-20µA	"	"									"	Q4	"		"		"		"
		27	"									"	"	-20µA	4.2V							"	Q5	"		"		"		"
		28	"									"	"			4.2V	-20µA					"	Q6	"		"		"		"
		29	"									"	"					-20µA	4.2V			"	Q7	"		"		"		"
		30	"									"	"							4.2V	-20µA	"	Q8	"		"		"		"
V_{OH5}		31	"	-5.2mA	4.2V							"	"									"	Q1	5.48		5.2		5.48		"
		32	"			4.2V	-5.2mA					"	"									"	Q2	"		"		"		"
		33	"					-5.2mA	4.2V			"	"									"	Q3	"		"		"		"
		34	"							4.2V	-5.2mA	"	"									"	Q4	"		"		"		"
		35	"									"	"	-5.2mA	4.2V							"	Q5	"		"		"		"
		36	"									"	"			4.2V	-5.2mA					"	Q6	"		"		"		"
		37	"									"	"					-5.2mA	4.2V			"	Q7	"		"		"		"
		38	"									"	"							4.2V	-5.2mA	"	Q8	"		"		"		"
V_{OL3}	3007	39	"	20µA	1.2V							"	"									"	Q1		0.1		0.1		0.1	"
		40	"			1.2V	20µA					"	"									"	Q2		"		"		"	"
		41	"					20µA	1.2V			"	"									"	Q3		"		"		"	"
		42	"							1.2V	20µA	"	"									"	Q4		"		"		"	"
		43	"									"	"	20µA	1.2V							"	Q5		"		"		"	"
		44	"									"	"			1.2V	20µA					"	Q6		"		"		"	"
		45	"									"	"					20µA	1.2V			"	Q7		"		"		"	"
		46	"									"	"							1.2V	20µA	"	Q8		"		"		"	"
V_{OL5}		47	"	5.2mA	1.2V							"	"									"	Q1		0.26		0.4		0.26	"
		48	"			1.2V	5.2mA					"	"									"	Q2		"		"		"	"
		49	"					5.2mA	1.2V			"	"									"	Q3		"		"		"	"
		50	"							1.2V	5.2mA	"	"									"	Q4		"		"		"	"
		51	"									"	"	5.2mA	1.2V							"	Q5		"		"		"	"
		52	"									"	"			1.2V	5.2mA					"	Q6		"		"		"	"
		53	"									"	"					5.2mA	1.2V			"	Q7		"		"		"	"
		54	"									"	"							1.2V	5.2mA	"	Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{CLR}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	55	4V	GND	4V	4V	GND	GND	4V	4V	GND	GND	3/	GND	4V	4V	GND	GND	4V	4V	GND	4V	Q1	-10	-120	-10	-120	-10	-120	mA
		56	"									"	"									"	Q2	"	"	"	"	"	"	"
		57	"									"	"									"	Q3	"	"	"	"	"	"	"
		58	"									"	"									"	Q4	"	"	"	"	"	"	"
		59	"									"	"									"	Q5	"	"	"	"	"	"	"
		60	"									"	"									"	Q6	"	"	"	"	"	"	"
		61	"									"	"									"	Q7	"	"	"	"	"	"	"
		62	"									"	"									"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	63	6V		GND	GND			GND	GND			GND		GND	GND			GND	GND		6V	CLR		.05		0.1			µA
		64	GND		6V	GND			"	"			"		"	"			"	"		"	D1		"		"			"
		65	"		GND	6V			"	"			"		"	"			"	"		"	D2		"		"			"
		66	"		"	GND			6V	6V			6V		"	"			"	"		"	D3		"		"			"
		67	"		"	"			GND	GND			GND		"	"			"	"		"	D4		"		"			"
		68	"		"	"			"	"			"		6V	6V			"	"		"	CLK		"		"			"
		69	"		"	"			"	"			"		GND	GND			"	"		"	D5		"		"			"
		70	"		"	"			"	"			"		"	"			"	"		"	D6		"		"			"
		71	"		"	"			"	"			"		"	"			6V	6V		"	D7		"		"			"
		72	"		"	"			"	"			"		"	"			GND	6V		"	D8		"		"			"
I_{IL}	3009	73	GND		6V	6V			6V	6V			6V		6V	6V			6V	"		"	CLR		-.05		-0.1			"
		74	6V		GND	6V			"	"			"		"	"			"	"		"	D1		"		"			"
		75	"		6V	GND			GND	GND			GND		"	"			"	"		"	D2		"		"			"
		76	"		"	6V			6V	6V			6V		"	"			"	"		"	D3		"		"			"
		77	"		"	"			"	"			"		"	"			"	"		"	D4		"		"			"
		78	"		"	"			"	"			"		"	"			"	"		"	CLK		"		"			"
		79	"		"	"			"	"			"		6V	6V			"	"		"	D5		"		"			"
		80	"		"	"			"	"			"		"	"			"	"		"	D6		"		"			"
		81	"		"	"			"	"			"		"	"			"	"		"	D7		"		"			"
		82	"		"	"			"	"			"		"	"			GND	GND		"	D8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	83	4/									GND										GND	CLR		15					pF
C_I		84			4/							"										"	D1		10					"
		85				4/						"										"	D2		"					"
		86							4/			"										"	D3		"					"
		87								4/		"										"	D4		"					"
C_C		88										"	4/									"	CLK		15					"
C_I		89										"			4/							"	D5		10					"
		90										"				4/						"	D6		"					"
		91										"							4/			"	D7		"					"
		92										"								4/			"	D8		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 9 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	93	B	L	A	A	L	L	A	A	L	"	A	L	A	A	L	L	A	A	L	4.5V	All outputs 7/ 8/							
		94	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		95	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		96	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		97	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		98	"	H	"	B	H	H	B	B	H	"	B	H	B	B	H	H	B	B	H	"									
		99	"	"	"	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"									
		100	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		101	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		102	"	L	"	B	L	L	B	B	L	"	B	L	B	B	L	L	B	B	L	"									
		103	"	"	"	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"									
		104	"	"	"	"	H	H	"	"	H	"	B	H	"	"	H	H	"	"	H	"									
		105	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		106	B	H	A	B	L	L	B	B	L	"	B	L	A	B	L	L	A	B	L	"									

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{CLR}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		107	4.5V	OUT	IN							GND	IN									4.5V	Q1	31		23		31		MHz
		108	"			IN	OUT		IN			"	"									"	Q2	"		"		"		"
		109	"					OUT		IN		"	"									"	Q3	"		"		"		"
		110	"								OUT	"	"									"	Q4	"		"		"		"
		111	"									"	"	OUT	IN							"	Q5	"		"		"		"
		112	"									"	"			IN	OUT					"	Q6	"		"		"		"
		113	"									"	"					OUT	IN			"	Q7	"		"		"		"
		114	"									"	"							IN	OUT	"	Q8	"		"		"		"
t_{PHL1}	3003 (Fig. 3)	115	IN	OUT	4.5V							GND	3/									4.5V	CLR to Q1		31		41		31	ns
		116	"			4.5V	OUT					"	"									"	CLR to Q2		"		"		"	"
		117	"					OUT	4.5V			"	"									"	CLR to Q3		"		"		"	"
		118	"							4.5V	OUT	"	"									"	CLR to Q4		"		"		"	"
		119	"									"	"	OUT	4.5V							"	CLR to Q5		"		"		"	"
		120	"									"	"			4.5V	OUT					"	CLR to Q6		"		"		"	"
		121	"									"	"					OUT	4.5V			"	CLR to Q7		"		"		"	"
		122	"									"	"							4.5V	OUT	"	CLR to Q8		"		"		"	"
t_{PLH2}		123	4.5V	OUT	IN							"	"									"	CLK to Q1		28		37		28	"
		124	"			IN	OUT					"	"									"	CLK to Q2		"		"		"	"
		125	"					OUT	IN			"	"									"	CLK to Q3		"		"		"	"
		126	"							IN	OUT	"	"									"	CLK to Q4		"		"		"	"
		127	"									"	"	OUT	IN							"	CLK to Q5		"		"		"	"
		128	"									"	"			IN	OUT					"	CLK to Q6		"		"		"	"
		129	"									"	"					OUT	IN			"	CLK to Q7		"		"		"	"
		130	"									"	"							IN	OUT	"	CLK to Q8		"		"		"	"
t_{PHL2}		131	"	OUT	IN							"	"									"	CLK to Q1		"		"		"	"
		132	"			IN	OUT					"	"									"	CLK to Q2		"		"		"	"
		133	"					OUT	IN			"	"									"	CLK to Q3		"		"		"	"
		134	"							IN	OUT	"	"									"	CLK to Q4		"		"		"	"
		135	"									"	"	OUT	IN							"	CLK to Q5		"		"		"	"
		136	"									"	"			IN	OUT					"	CLK to Q6		"		"		"	"
		137	"									"	"					OUT	IN			"	CLK to Q7		"		"		"	"
		138	"									"	"							IN	OUT	"	CLK to Q8		"		"		"	"
t_{THL}	3004	139	"	OUT	IN	4.5V			4.5V	4.5V		"	"		4.5V	4.5V			4.5V	4.5V		"	Q1		15		20		15	ns
		140	"		4.5V	IN	OUT		4.5V	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		141	"		"	4.5V		OUT	IN			"	"		"				"			"	Q3		"		"		"	"
		142	"		"	"			4.5V	IN	OUT	"	"		"	"			"	"		"	Q4		"		"		"	"
		143	"		"	"			"	4.5V		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		144	"		"	"			"	"		"	"		"	IN	OUT		"	"		"	Q6		"		"		"	"
		145	"		"	"			"	"		"	"		"	4.5V		OUT	IN	"		"	Q7		"		"		"	"
		146	"		"	"			"	"		"	"		"	"			4.5V	IN	OUT	"	Q8		"		"		"	"
t_{TLH}		147	"	OUT	IN	GND			GND	GND		"	"		GND	GND			GND	GND		"	Q1		"		"		"	"
		148	"		GND	IN	OUT		GND	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		149	"		"	GND		OUT	IN			"	"		"				"			"	Q3		"		"		"	"
		150	"		"	"			GND	IN	OUT	"	"		"	"			"	"		"	Q4		"		"		"	"
		151	"		"	"			"	GND		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		152	"		"	"			"	"		"	"		"	IN	OUT		"	"		"	Q6		"		"		"	"
		153	"		"	"			"	"		"	"		"	GND		OUT	IN	"		"	Q7		"		"		"	"
		154	"		"	"			"	"		"	"		"	"			GND	IN	OUT	"	Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA		1mA	1mA			1mA	1mA		1/	1mA		1mA	1mA			1mA	1mA		GND	$\overline{OE}$	2/	1.5					V
		2										"										"	D1		"					"
		3										"										"	D2		"					"
		4										"										"	D3		"					"
		5										"										"	D4		"					"
		6										"										"	CLK		"					"
		7										"										"	D5		"					"
		8										"										"	D6		"					"
		9										"										"	D7		"					"
		10										"										"	D8		"					"
V_{IC}(neg)		11	-1mA		-1mA	-1mA			-1mA	-1mA		GND	-1mA		-1mA	-1mA			-1mA	-1mA		1/	$\overline{OE}$		-1.5					"
		12										"										"	D1		"					"
		13										"										"	D2		"					"
		14										"										"	D3		"					"
		15										"										"	D4		"					"
		16										"										"	CLK		"					"
		17										"										"	D5		"					"
		18										"										"	D6		"					"
		19										"										"	D7		"					"
		20										"										"	D8		"					"
I_{CCH}	3005	21	GND		6V	6V			6V	6V		"	3/		6V	6V			6V	6V		6V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	"		0.2		8.0			"
I_{CCZ}		23	6V		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	"		0.1		4.0			"
V_{OH3}	3006	24	1.2V	-20µA	4.2V	4.2V	-20µA	-20µA	4.2V	4.2V	-20µA	"	"	-20µA	4.2V	4.2V	-20µA	-20µA	4.2V	4.2V	-20µA	"	Q1	5.95		5.95		5.95		V
		25	"									"	"									"	Q2	"		"		"		"
		26	"									"	"									"	Q3	"		"		"		"
		27	"									"	"									"	Q4	"		"		"		"
		28	"									"	"									"	Q5	"		"		"		"
		29	"									"	"									"	Q6	"		"		"		"
		30	"									"	"									"	Q7	"		"		"		"
		31	"									"	"									"	Q8	"		"		"		"
V_{OH5}		32	"	-5.2mA	4.2V	4.2V	-5.2mA	-5.2mA	4.2V	4.2V	-5.2mA	"	"	-5.2mA	4.2V	4.2V	-5.2mA	-5.2mA	4.2V	4.2V	-5.2mA	"	Q1	5.48		5.2		5.48		"
		33	"									"	"									"	Q2	"		"		"		"
		34	"									"	"									"	Q3	"		"		"		"
		35	"									"	"									"	Q4	"		"		"		"
		36	"									"	"									"	Q5	"		"		"		"
		37	"									"	"									"	Q6	"		"		"		"
		38	"									"	"									"	Q7	"		"		"		"
		39	"									"	"									"	Q8	"		"		"		"
V_{OL3}	3007	40	"	20µA	1.2V	1.2V	20µA	20µA	1.2V	1.2V	20µA	"	"	20µA	1.2V	1.2V	20µA	20µA	1.2V	1.2V	20µA	"	Q1		0.1		0.1		0.1	"
		41	"									"	"									"	Q2		"		"		"	"
		42	"									"	"									"	Q3		"		"		"	"
		43	"									"	"									"	Q4		"		"		"	"
		44	"									"	"									"	Q5		"		"		"	"
		45	"									"	"									"	Q6		"		"		"	"
		46	"									"	"									"	Q7		"		"		"	"
		47	"									"	"									"	Q8		"		"		"	"
V_{OL5}		48	"	5.2mA	1.2V	1.2V	5.2mA	5.2mA	1.2V	1.2V	5.2mA	"	"	5.2mA	1.2V	1.2V	5.2mA	5.2mA	1.2V	1.2V	5.2mA	"	Q1		0.26		0.4		0.26	"
		49	"									"	"									"	Q2		"		"		"	"
		50	"									"	"									"	Q3		"		"		"	"
		51	"									"	"									"	Q4		"		"		"	"
		52	"									"	"									"	Q5		"		"		"	"
		53	"									"	"									"	Q6		"		"		"	"
		54	"									"	"									"	Q7		"		"		"	"
		55	"									"	"									"	Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	GND	4V	4V	GND	GND	4V	4V	GND	GND	3/	GND	4V	4V	GND	GND	4V	4V	GND	4V	Q1	-10	-135	-10	-135	-10	-135	mA
		57	"									"	"									"	Q2	"	"	"	"	"	"	"
		58	"									"	"									"	Q3	"	"	"	"	"	"	"
		59	"									"	"									"	Q4	"	"	"	"	"	"	"
		60	"									"	"									"	Q5	"	"	"	"	"	"	"
		61	"									"	"									"	Q6	"	"	"	"	"	"	"
		62	"									"	"									"	Q7	"	"	"	"	"	"	"
		63	"									"	"									"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	64	6V		GND	GND			GND	GND		"	GND		GND	GND			GND	GND		6V	$\overline{OE}$		.05		0.1			µA
		65	GND		6V	GND			"	"		"	"		"	"			"	"		"	D1		"		"			"
		66	"		GND	6V			"	"		"	"		"	"			"	"		"	D2		"		"			"
		67	"		"	GND			6V	6V		"	"		"	"			"	"		"	D3		"		"			"
		68	"		"	"			GND	GND		"	"		"	"			"	"		"	D4		"		"			"
		69	"		"	"			"	"		"	6V		"	"			"	"		"	CLK		"		"			"
		70	"		"	"			"	"		"	GND		"	"			"	"		"	D5		"		"			"
		71	"		"	"			"	"		"	"		6V	6V			"	"		"	D6		"		"			"
		72	"		"	"			"	"		"	"		GND	GND			6V	"		"	D7		"		"			"
		73	"		"	"			"	"		"	"		"	"			GND	6V		"	D8		"		"			"
I_{IL}	3009	74	GND		6V	6V			6V	6V		"	6V		6V	6V			6V	"		"	$\overline{OE}$		-.05		-0.1			"
		75	6V		GND	6V			"	"		"	"		"	"			"	"		"	D1		"		"			"
		76	"		6V	GND			"	"		"	"		"	"			"	"		"	D2		"		"			"
		77	"		"	6V			GND	GND		"	"		"	"			"	"		"	D3		"		"			"
		78	"		"	"			6V	6V		"	"		"	"			"	"		"	D4		"		"			"
		79	"		"	"			"	"		"	6V		"	"			"	"		"	CLK		"		"			"
		80	"		"	"			"	"		"	"		GND	GND			"	"		"	D5		"		"			"
		81	"		"	"			"	"		"	"		6V	6V			"	"		"	D6		"		"			"
		82	"		"	"			"	"		"	"		"	"			GND	GND		"	D7		"		"			"
		83	"		"	"			"	"		"	"		"	"			6V	"		"	D8		"		"			"
I_{OZH}		84	4.2V	6.0V	1.2V	1.2V	6.0V	6.0V	1.2V	1.2V	6.0V	"	3/	6.0V	1.2V	1.2V	6.0V	6.0V	1.2V	1.2V	6.0V	"	Q1		0.2		2.0			"
		85	"		"	"			"	"		"	"		"	"			"	"		"	Q2		"		"			"
		86	"		"	"			"	"		"	"		"	"			"	"		"	Q3		"		"			"
		87	"		"	"			"	"		"	"		"	"			"	"		"	Q4		"		"			"
		88	"		"	"			"	"		"	"		"	"			"	"		"	Q5		"		"			"
		89	"		"	"			"	"		"	"		"	"			"	"		"	Q6		"		"			"
		90	"		"	"			"	"		"	"		"	"			"	"		"	Q7		"		"			"
		91	"		"	"			"	"		"	"		"	"			"	"		"	Q8		"		"			"
I_{OZL}		92	"	GND	4.2V	4.2V	GND	GND	4.2V	4.2V	GND	"	"	GND	4.2V	4.2V	GND	GND	4.2V	4.2V	GND	"	Q1		-0.2		-2.0			"
		93	"		"	"			"	"		"	"		"	"			"	"		"	Q2		"		"			"
		94	"		"	"			"	"		"	"		"	"			"	"		"	Q3		"		"			"
		95	"		"	"			"	"		"	"		"	"			"	"		"	Q4		"		"			"
		96	"		"	"			"	"		"	"		"	"			"	"		"	Q5		"		"			"
		97	"		"	"			"	"		"	"		"	"			"	"		"	Q6		"		"			"
		98	"		"	"			"	"		"	"		"	"			"	"		"	Q7		"		"			"
		99	"		"	"			"	"		"	"		"	"			"	"		"	Q8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	100	4/									GND										GND	$\overline{OE}$		15					pF
C_I		101			4/							"										"	D1		10					"
		102				4/						"										"	D2		"					"
		103							4/			"										"	D3		"					"
		104								4/		"										"	D4		"					"
C_C		105										"	4/									"	CLK		15					"
C_I		106										"			4/							"	D5		10					"
		107										"				4/						"	D6		"					"
		108										"							4/			"	D7		"					"
		109										"								4/			"	D8		"					"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
C_O	3012	110	6V	4/								GND										6V	Q1		20				pF
		111	"				4/					"										"	Q2		"				"
		112	"					4/				"										"	Q3		"				"
		113	"								4/	"										"	Q4		"				"
		114	"									"		4/								"	Q5		"				"
		115	"									"					4/					"	Q6		"				"
		116	"									"						4/				"	Q7		"				"
		117	"									"									4/	"	Q8		"				"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	118	B	X	A	A	X	X	A	A	X	"	B	X	A	A	X	X	A	A	X	4.5V	All outputs 7/ 8/							
		119	"	H	B	B	H	H	B	B	H	"	A	H	B	B	H	H	B	B	H	"								
		120	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		121	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"									
		122	"	L	"	"	L	L	"	"	L	"	B	L	B	B	L	L	"	"	L	"									
		123	"	L	"	"	L	L	"	"	L	"	A	L	B	B	L	L	"	"	L	"									
		124	"	H	A	A	H	H	A	A	H	"	B	H	A	A	H	H	A	A	H	"									
																								Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
																								Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		125	GND	OUT	IN		OUT		IN		OUT	GND	IN		IN		OUT		IN			4.5V	Q1	33		25		33		MHz
		126	"			IN		OUT		IN		"	"			IN		OUT				"	Q2	"		"		"		"
		127	"						OUT		IN	"	"		IN		OUT					"	Q3	"		"		"		"
		128	"				IN					"	"	OUT					IN			"	Q4	"		"		"		"
		129	"					OUT				"	"		IN			OUT					"	Q5	"		"		"		"
		130	"							IN		"	"			IN							"	Q6	"		"		"		"
		131	"									"	"						OUT	IN			"	Q7	"		"		"		"
		132	"									"	"								IN	OUT	"	Q8	"		"		"		"
t_{PHL2}	3003 (Fig. 3)	133	"	OUT	IN	4.5V	OUT		4.5V	4.5V		"	"		4.5V	4.5V			4.5V	4.5V			"	CLK to Q1		32		42		32	ns
		134	"		4.5V	IN			4.5V	"		"	"		"	"			"	"			"	CLK to Q2		"		"		"	"
		135	"		"	4.5V			IN			"	"		"	"			"	"			"	CLK to Q3		"		"		"	"
		136	"		"	"		OUT	4.5V	IN		"	"		"	"			"	"			"	CLK to Q4		"		"		"	"
		137	"		"	"			"	4.5V	OUT	"	"	OUT	IN	"			"	"			"	CLK to Q5		"		"		"	"
		138	"		"	"			"	"		"	"		4.5V	IN	OUT		"	"			"	CLK to Q6		"		"		"	"
		139	"		"	"			"	"		"	"		"	4.5V		OUT	IN	"			"	CLK to Q7		"		"		"	"
		140	"		"	"			"	"		"	"		"	4.5V			4.5V	IN	OUT		"	CLK to Q8		"		"		"	"
t_{PLH2}		141	"	OUT	IN	GND	OUT		GND	GND		"	"		GND	GND			GND	GND			"	CLK to Q1		"		"		"	"
		142	"		GND	IN			GND	"		"	"		"	"			"	"			"	CLK to Q2		"		"		"	"
		143	"		"	GND			IN			"	"		"	"			"	"			"	CLK to Q3		"		"		"	"
		144	"		"	"		OUT	GND	IN		"	"		"	"			"	"			"	CLK to Q4		"		"		"	"
		145	"		"	"			"	GND	OUT	"	"	OUT	IN	"			"	"			"	CLK to Q5		"		"		"	"
		146	"		"	"			"	"		"	"		GND	IN	OUT		"	"			"	CLK to Q6		"		"		"	"
		147	"		"	"			"	"		"	"		"	GND		OUT	IN	"			"	CLK to Q7		"		"		"	"
		148	"		"	"			"	"		"	"		"	GND			GND	IN	OUT		"	CLK to Q8		"		"		"	"
t_{PZH}		149	IN	OUT	4.5V		OUT		4.5V			"			4.5V								"	$\overline{OE}$ to Q1		26		35		26	"
		150	"			4.5V						"			"								"	$\overline{OE}$ to Q2		"		"		"	"
		151	"		"					4.5V		"			"								"	$\overline{OE}$ to Q3		"		"		"	"
		152	"		"	"		OUT			4.5V	"			"								"	$\overline{OE}$ to Q4		"		"		"	"
		153	"		"	"						"		OUT	"								"	$\overline{OE}$ to Q5		"		"		"	"
		154	"		"	"						"			"	4.5V	OUT						"	$\overline{OE}$ to Q6		"		"		"	"
		155	"		"	"						"			"			OUT	4.5V				"	$\overline{OE}$ to Q7		"		"		"	"
		156	"		"	"						"			"					4.5V	OUT		"	$\overline{OE}$ to Q8		"		"		"	"

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TABLE III. Group A inspection for device type 02 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZL}	3003 (Fig. 3)	157	IN	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND		OUT	GND	GND	OUT	OUT	GND	GND	OUT	4.5V	$\overline{OE}$ to Q1		26		35		26	ns
		158	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		159	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		160	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		161	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		162	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		163	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		164	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{PHZ}		165	"	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"		OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{OE}$ to Q1		"		"		"	"
		166	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		167	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		168	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		169	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		170	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		171	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		172	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{PLZ}		173	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"		OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to Q1		"		"		"	"
		174	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		175	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		176	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		177	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		178	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		179	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		180	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{THL}	3004	181	GND	OUT	IN	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	Q1		12		16		12	ns
		182	"		4.5V	IN			4.5V	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		183	"		"	4.5V			IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		184	"		"	"			4.5V	IN		"	"		"	"			"	"		"	Q4		"		"		"	"
		185	"		"	"			"	4.5V		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		186	"		"	"			"	"		"	"		4.5V	IN	OUT		"	"		"	Q6		"		"		"	"
		187	"		"	"			"	"		"	"		"	4.5V		OUT	IN	"		"	Q7		"		"		"	"
		188	"		"	"			"	"		"	"		"	4.5V			4.5V	IN	OUT	"	Q8		"		"		"	"
t_{TLH}		189	"	OUT	IN	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	Q1		"		"		"	"
		190	"		GND	IN			GND	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		191	"		"	GND			IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		192	"		"	"			GND	IN		"	"		"	"			"	"		"	Q4		"		"		"	"
		193	"		"	"			"	GND		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		194	"		"	"			"	"		"	"		GND	IN	OUT		"	"		"	Q6		"		"		"	"
		195	"		"	"			"	"		"	"		"	GND		OUT	IN	"		"	Q7		"		"		"	"
		196	"		"	"			"	"		"	"		"	GND			GND	IN	OUT	"	Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{CE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA									1/										GND	$\overline{CE}$	2/	1.5					V
		2			1mA							"										"	D1		"					"
		3				1mA						"										"	D2		"					"
		4							1mA			"										"	D3		"					"
		5								1mA		"										"	D4		"					"
		6										"	1mA									"	CLK		"					"
		7										"			1mA							"	D5		"					"
		8										"				1mA						"	D6		"					"
		9										"							1mA			"	D7		"					"
		10										"								1mA			"	D8		"					"
V_{IC}(neg)		11	-1mA									GND										1/	$\overline{CE}$		-1.5					"
		12			-1mA							"										"	D1		"					"
		13				-1mA						"										"	D2		"					"
		14							-1mA			"										"	D3		"					"
		15								-1mA		"										"	D4		"					"
		16										"	-1mA									"	CLK		"					"
		17										"			-1mA							"	D5		"					"
		18										"				-1mA						"	D6		"					"
		19										"							-1mA			"	D7		"					"
		20										"								-1mA			"	D8		"					"
I_{CCH}	3005	21	6V		6V	6V			6V	6V		"	3/		6V	6V			6V	6V		6V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	V_{CC}		0.2		8.0			µA
V_{OH3}	3006	23	1.2V	-20µA	4.2V																		Q1	5.95		5.95		5.95		V
		24	"	"	"	4.2V	-20µA																Q2	"		"		"		"
		25	"	"	"	"	"	-20µA	4.2V														Q3	"		"		"		"
		26	"	"	"	"	"	"	"	4.2V	-20µA												Q4	"		"		"		"
		27	"	"	"	"	"	"	"	"	"			-20µA	4.2V								Q5	"		"		"		"
		28	"	"	"	"	"	"	"	"	"			"	"	4.2V	-20µA						Q6	"		"		"		"
		29	"	"	"	"	"	"	"	"	"			"	"	"	"	-20µA	4.2V				Q7	"		"		"		"
		30	"	"	"	"	"	"	"	"	"			"	"	"	"	"	"	4.2V	-20µA		Q8	"		"		"		"
V_{OH5}		31	"	-5.2mA	4.2V																		Q1	5.48		5.2		5.48		"
		32	"	"	"	4.2V	-5.2mA																Q2	"		"		"		"
		33	"	"	"	"	"	-5.2mA	4.2V														Q3	"		"		"		"
		34	"	"	"	"	"	"	"	4.2V	-5.2mA												Q4	"		"		"		"
		35	"	"	"	"	"	"	"	"	"			-5.2mA	4.2V								Q5	"		"		"		"
		36	"	"	"	"	"	"	"	"	"			"	"	4.2V	-5.2mA						Q6	"		"		"		"
		37	"	"	"	"	"	"	"	"	"			"	"	"	"	-5.2mA	4.2V				Q7	"		"		"		"
		38	"	"	"	"	"	"	"	"	"			"	"	"	"	"	"	4.2V	-5.2mA		Q8	"		"		"		"
V_{OL3}	3007	39	"	20µA	1.2V																		Q1		0.1		0.1		0.1	"
		40	"	"	"	1.2V	20µA																Q2		"		"		"	"
		41	"	"	"	"	"	20µA	1.2V														Q3		"		"		"	"
		42	"	"	"	"	"	"	"	1.2V	20µA												Q4		"		"		"	"
		43	"	"	"	"	"	"	"	"	"			20µA	1.2V								Q5		"		"		"	"
		44	"	"	"	"	"	"	"	"	"			"	"	1.2V	20µA						Q6		"		"		"	"
		45	"	"	"	"	"	"	"	"	"			"	"	"	"	20µA	1.2V				Q7		"		"		"	"
		46	"	"	"	"	"	"	"	"	"			"	"	"	"	"	"	1.2V	20µA		Q8		"		"		"	"
V_{OL5}		47	"	5.2mA	1.2V																		Q1		0.26		0.4		0.26	"
		48	"	"	"	1.2V	5.2mA																Q2		"		"		"	"
		49	"	"	"	"	"	5.2mA	1.2V														Q3		"		"		"	"
		50	"	"	"	"	"	"	"	1.2V	5.2mA												Q4		"		"		"	"
		51	"	"	"	"	"	"	"	"	"			5.2mA	1.2V								Q5		"		"		"	"
		52	"	"	"	"	"	"	"	"	"			"	"	1.2V	5.2mA						Q6		"		"		"	"
		53	"	"	"	"	"	"	"	"	"			"	"	"	"	5.2mA	1.2V				Q7		"		"		"	"
		54	"	"	"	"	"	"	"	"	"			"	"	"	"	"	"	1.2V	5.2mA		Q8		"		"		"	"

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{CE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	55	GND	GND	4V		GND					GND	3/									4V	Q1	-10	-120	-10	-120	-10	-120	mA
		56	"			4V		GND	4V		GND	"	"									"	Q2	"	"	"	"	"	"	"
		57	"							4V		"	"	GND	4V							"	Q3	"	"	"	"	"	"	"
		58	"									"	"			4V	GND					"	Q4	"	"	"	"	"	"	"
		59	"									"	"					GND				"	Q5	"	"	"	"	"	"	"
		60	"									"	"						4V			"	Q6	"	"	"	"	"	"	"
		61	"									"	"							4V	GND	"	Q7	"	"	"	"	"	"	"
		62	"									"	"									"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	63	6V		GND	GND			GND	GND		"	GND		GND	GND			GND	GND		6V	CE		.05		0.1			µA
		64	GND		6V	GND			"	"		"	"		"	"			"	"		"	D1		"		"			"
		65	"		GND	6V			"	"		"	"		"	"			"	"		"	D2		"		"			"
		66	"		"	GND			6V	6V		"	"		"	"			"	"		"	D3		"		"			"
		67	"		"	"			GND	GND		"	"		"	"			"	"		"	D4		"		"			"
		68	"		"	"			"	"		"	6V		"	"			"	"		"	CLK		"		"			"
		69	"		"	"			"	"		"	GND		6V	"			"	"		"	D5		"		"			"
		70	"		"	"			"	"		"	"		GND	6V			"	"		"	D6		"		"			"
		71	"		"	"			"	"		"	"		"	GND			6V	"		"	D7		"		"			"
		72	"		"	"			"	"		"	"		"	"			GND	6V		"	D8		"		"			"
I_{IL}	3009	73	GND		6V	6V			6V	6V		"	6V		6V	6V			6V	"		"	CE		-.05		-0.1			"
		74	6V		GND	6V			"	"		"	"		"	"			"	"		"	D1		"		"			"
		75	"		6V	GND			GND	GND		"	"		"	"			"	"		"	D2		"		"			"
		76	"		"	6V			6V	6V		"	"		"	"			"	"		"	D3		"		"			"
		77	"		"	"			"	"		"	"		"	"			"	"		"	D4		"		"			"
		78	"		"	"			"	"		"	6V		"	"			"	"		"	CLK		"		"			"
		79	"		"	"			"	"		"	GND		6V	"			"	"		"	D5		"		"			"
		80	"		"	"			"	"		"	"		GND	6V			"	"		"	D6		"		"			"
		81	"		"	"			"	"		"	"		"	GND			6V	"		"	D7		"		"			"
		82	"		"	"			"	"		"	"		"	"			GND	6V		"	D8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	83	4/									GND										GND	CE		15					pF
C_I		84			4/							"										"	D1		10					"
		85				4/						"										"	D2		"					"
		86							4/			"										"	D3		"					"
		87								4/		"										"	D4		"					"
C_C		88										"	4/									"	CLK		15					"
C_I		89										"			4/							"	D5		10					"
		90										"				4/						"	D6		"					"
		91										"							4/			"	D7		"					"
		92										"								4/			"	D8		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	93	B	X	B	B	X	X	B	B	X	"	B	X	B	B	X	X	B	B	X	4.5V	All outputs 7/ 8/							
		94	"	L	B	B	L	L	B	B	L	"	A	L	B	B	L	L	B	B	L	"								
		95	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		96	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		97	"	H	B	B	H	H	B	B	H	"	B	H	B	B	H	H	B	B	H	"								
		98	A	"	B	B	"	"	B	B	"	"	B	"	B	B	"	"	B	B	"	"								
		99	"	"	B	B	"	"	B	B	"	"	B	"	B	B	"	"	B	B	"	"								
		100	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		101	B	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		102	"	"	B	B	"	"	B	B	"	"	B	"	B	B	"	"	B	B	"	"								
		103	"	L	A	A	L	L	A	A	L	"	A	L	A	A	L	L	A	A	L	"								
		104	A	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		105	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		106	"	"	B	B	"	"	B	B	"	"	B	"	B	B	"	"	B	B	"	"								
		107	B	"	B	B	"	"	B	B	"	"	B	"	B	B	"	"	B	B	"	"								
		108	"	"	A	A	"	"	A	A	"	"	A	"	A	A	"	"	A	A	"	"								
		109	"	H	A	A	H	H	A	A	H	"	A	H	A	A	H	H	A	A	H	"								

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TABLE III. Group A inspection for device type 03 – Continued.

Symbol	MIL-STD-883 method	Cases R, S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{CE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		110	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	Q1	28		21		28		MHz
		111	"									"	"									"	Q2	"		"		"		"
		112	"									"	"									"	Q3	"		"		"		"
		113	"									"	"									"	Q4	"		"		"		"
		114	"									"	"									"	Q5	"		"		"		"
		115	"									"	"									"	Q6	"		"		"		"
		116	"									"	"									"	Q7	"		"		"		"
		117	"									"	"									"	Q8	"		"		"		"
t_{PLH2}	3003 (Fig. 3)	118	GND	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND	IN	OUT	IN	IN	OUT	OUT	IN	IN	OUT	4.5V	CLK to Q1		28		37		28	ns
		119	"									"	"									"	CLK to Q2		"		"		"	"
		120	"									"	"									"	CLK to Q3		"		"		"	"
		121	"									"	"									"	CLK to Q4		"		"		"	"
		122	"									"	"									"	CLK to Q5		"		"		"	"
		123	"									"	"									"	CLK to Q6		"		"		"	"
		124	"									"	"									"	CLK to Q7		"		"		"	"
		125	"									"	"									"	CLK to Q8		"		"		"	"
t_{PHL2}		126	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	"	OUT	IN	IN	OUT	OUT	IN	IN	OUT	"	CLK to Q1		"		"		"	"
		127	"									"	"									"	CLK to Q2		"		"		"	"
		128	"									"	"									"	CLK to Q3		"		"		"	"
		129	"									"	"									"	CLK to Q4		"		"		"	"
		130	"									"	"									"	CLK to Q5		"		"		"	"
		131	"									"	"									"	CLK to Q6		"		"		"	"
		132	"									"	"									"	CLK to Q7		"		"		"	"
		133	"									"	"									"	CLK to Q8		"		"		"	"
t_{THL}	3004	134	4.5V	OUT	IN	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	"	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	Q1		15		20		15	ns
		135	"		4.5V	IN			4.5V	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		136	"		"	4.5V			IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		137	"		"	"			4.5V	IN		"	"		"	"			"	"		"	Q4		"		"		"	"
		138	"		"	"			"	4.5V		"	"		IN	"			"	"		"	Q5		"		"		"	"
		139	"		"	"			"	"		"	"		4.5V	IN			"	"		"	Q6		"		"		"	"
		140	"		"	"			"	"		"	"		"	4.5V			"	"		"	Q7		"		"		"	"
		141	"		"	"			"	"		"	"		"	4.5V			IN	IN		"	Q8		"		"		"	"
t_{TLH}		142	"	OUT	IN	GND	OUT	OUT	GND	GND	OUT	"	"	OUT	GND	GND	OUT	OUT	GND	GND		"	Q1		"		"		"	"
		143	"		GND	IN			GND	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		144	"		"	GND			IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		145	"		"	"			GND	IN		"	"		"	"			"	"		"	Q4		"		"		"	"
		146	"		"	"			"	GND		"	"		IN	"			"	"		"	Q5		"		"		"	"
		147	"		"	"			"	"		"	"		GND	IN			"	"		"	Q6		"		"		"	"
		148	"		"	"			"	"		"	"		"	GND			IN	"		"	Q7		"		"		"	"
		149	"		"	"			"	"		"	"		"	GND			GND	IN		"	Q8		"		"		"	"

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Cases R,S, 2, X	Terminal conditions 1/																			Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	Q8	Q7	Q6	Q5	Q4	Q3	Q2	Q1	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA									1/										GND	$\overline{OE}$	2/	1.5					V
		2		1mA								"										"	D1		"					"
		3			1mA							"										"	D2		"					"
		4				1mA						"										"	D3		"					"
		5					1mA					"										"	D4		"					"
		6						1mA				"										"	D5		"					"
		7							1mA			"										"	D6		"					"
		8								1mA		"										"	D7		"					"
		9									1mA	"										"	D8		"					"
		10										"	1mA									"	CLK		"					"
V_{IC}(neg)		11	-1mA									GND										1/	$\overline{OE}$		-1.5					"
		12		-1mA								"										"	D1		"					"
		13			-1mA							"										"	D2		"					"
		14				-1mA						"										"	D3		"					"
		15					-1mA					"										"	D4		"					"
		16						-1mA				"										"	D5		"					"
		17							-1mA			"										"	D6		"					"
		18								-1mA		"										"	D7		"					"
		19									-1mA	"										"	D8		"					"
		20										"	-1mA									"	CLK		"					"
I_{CCH}	3005	21	GND	6V	6V	6V	6V	6V	6V	6V	6V	"	3/									6V	V_{CC}		0.2		8.0			μA
I_{CCL}		22	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	"									"	"		0.2		8.0			"
I_{CCZ}		23	6V	GND	GND	GND	GND	GND	GND	GND	GND	"	"									"	"		.1		4			"
V_{OH3}	3006	24	1.2V	4.2V								"	"								-20μA	"	Q1	5.95		5.95		5.95		V
		25	"		4.2V							"	"							-20μA	"	"	Q2	"		"		"		"
		26	"			4.2V						"	"						-20μA		"	"	Q3	"		"		"		"
		27	"				4.2V					"	"					-20μA			"	"	Q4	"		"		"		"
		28	"					4.2V				"	"				-20μA				"	"	Q5	"		"		"		"
		29	"						4.2V			"	"			-20μA					"	"	Q6	"		"		"		"
		30	"							4.2V		"	"		-20μA						"	"	Q7	"		"		"		"
		31	"								4.2V	"	"	-20μA							"	"	Q8	"		"		"		"
V_{OH5}		32	"	4.2V								"	"							-5.2mA	"	Q1	5.48		5.2		5.48		"
		33	"		4.2V							"	"						-5.2mA		"	"	Q2	"		"		"		"
		34	"			4.2V						"	"					-5.2mA			"	"	Q3	"		"		"		"
		35	"				4.2V					"	"				-5.2mA				"	"	Q4	"		"		"		"
		36	"					4.2V				"	"			-5.2mA					"	"	Q5	"		"		"		"
		37	"						4.2V			"	"		-5.2mA						"	"	Q6	"		"		"		"
		38	"							4.2V		"	"								"	"	Q7	"		"		"		"
		39	"								4.2V	"	"	-5.2mA							"	"	Q8	"		"		"		"
V_{OL3}	3007	40	"	1.2V								"	"							20μA	"	Q1		0.1		0.1		0.1	"
		41	"		1.2V							"	"						20μA		"	"	Q2		"		"		"	"
		42	"			1.2V						"	"					20μA			"	"	Q3		"		"		"	"
		43	"				1.2V					"	"				20μA				"	"	Q4		"		"		"	"
		44	"					1.2V				"	"			20μA					"	"	Q5		"		"		"	"
		45	"						1.2V			"	"		20μA						"	"	Q6		"		"		"	"
		46	"							1.2V		"	"								"	"	Q7		"		"		"	"
		47	"								1.2V	"	"	20μA							"	"	Q8		"		"		"	"
V_{OL5}		48	"	1.2V								"	"							5.2mA	"	Q1		0.26		0.4		0.26	"
		49	"		1.2V							"	"						5.2mA		"	"	Q2		"		"		"	"
		50	"			1.2V						"	"					5.2mA			"	"	Q3		"		"		"	"
		51	"				1.2V					"	"				5.2mA				"	"	Q4		"		"		"	"
		52	"					1.2V				"	"			5.2mA					"	"	Q5		"		"		"	"
		53	"						1.2V			"	"		5.2mA						"	"	Q6		"		"		"	"
		54	"							1.2V		"	"								"	"	Q7		"		"		"	"
		55	"								1.2V	"	"	5.2mA							"	"	Q8		"		"		"	"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2, X	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	Q8	Q7	Q6	Q5	Q4	Q3	Q2	Q1	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	4V								GND	3/									4V	Q1	-10	-135	-10	-135	-10	-135	mA
		57	"		4V							"	"							GND		"	Q2	"	"	"	"	"	"	"
		58	"			4V						"	"						GND		GND	"	Q3	"	"	"	"	"	"	"
		59	"				4V					"	"					GND				"	Q4	"	"	"	"	"	"	"
		60	"					4V				"	"				GND					"	Q5	"	"	"	"	"	"	"
		61	"						4V			"	"			GND						"	Q6	"	"	"	"	"	"	"
		62	"							4V		"	"		GND							"	Q7	"	"	"	"	"	"	"
		63	"								4V	"	"	GND								"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	64	6V	GND	GND	GND	GND	GND	GND	GND	GND	"	GND									6V	$\overline{OE}$		.05		0.1			µA
		65	GND	6V	GND	"	"	"	"	"	"	"	"									"	D1		"		"			"
		66	"	GND	6V	GND	"	"	"	"	"	"	"									"	D2		"		"			"
		67	"	"	GND	"	6V	"	"	"	"	"	"									"	D3		"		"			"
		68	"	"	"	6V	GND	"	"	"	"	"	"									"	D4		"		"			"
		69	"	"	"	"	"	6V	"	"	"	"	"									"	D5		"		"			"
		70	"	"	"	"	6V	GND	6V	"	"	"	"									"	D6		"		"			"
		71	"	"	"	"	GND	"	GND	6V	"	"	"									"	D7		"		"			"
		72	"	"	"	"	"	"	"	GND	6V	"	"									"	D8		"		"			"
		73	"	"	"	"	"	"	"	"	GND	"	6V									"	CLK		"		"			"
I_{IL}	3009	74	GND	6V	6V	6V	6V	6V	6V	6V	6V	"	6V									"	$\overline{OE}$		-.05		-0.1			"
		75	6V	GND	6V	"	"	"	"	"	"	"	"									"	D1		"		"			"
		76	"	6V	GND	GND	"	"	"	"	"	"	"									"	D2		"		"			"
		77	"	"	6V	"	GND	"	"	"	"	"	"									"	D3		"		"			"
		78	"	"	"	GND	6V	GND	"	"	"	"	"									"	D4		"		"			"
		79	"	"	"	"	"	6V	GND	"	"	"	"									"	D5		"		"			"
		80	"	"	"	"	GND	"	6V	GND	"	"	"									"	D6		"		"			"
		81	"	"	"	"	"	GND	"	6V	GND	"	"									"	D7		"		"			"
		82	"	"	"	"	"	"	"	"	6V	"	"									"	D8		"		"			"
		83	"	"	"	"	"	"	"	"	"	"	GND									"	CLK		"		"			"
I_{OZH}		84	4.2V	1.2V								"	3/									"	Q1		0.2		2.0			"
		85	"		1.2V							"	"							6.0V	6.0V	"	Q2		"		"			"
		86	"			1.2V						"	"						6.0V			"	Q3		"		"			"
		87	"				1.2V					"	"					6.0V				"	Q4		"		"			"
		88	"					1.2V				"	"				6.0V					"	Q5		"		"			"
		89	"						1.2V			"	"			6.0V						"	Q6		"		"			"
		90	"							1.2V		"	"		6.0V							"	Q7		"		"			"
		91	"								1.2V	"	"	6.0V								"	Q8		"		"			"
I_{OZL}		92	"	4.2V								"	"									"	Q1		-0.2		-2.0			"
		93	"		4.2V							"	"							GND	GND	"	Q2		"		"			"
		94	"			4.2V						"	"						GND			"	Q3		"		"			"
		95	"				4.2V					"	"					GND				"	Q4		"		"			"
		96	"					4.2V				"	"				GND					"	Q5		"		"			"
		97	"						4.2V			"	"			GND						"	Q6		"		"			"
		98	"							4.2V		"	"		GND							"	Q7		"		"			"
		99	"								4.2V	"	"	GND								"	Q8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	100	4/									GND										GND	$\overline{OE}$		15					pF
C_I		101		4/								"										"	D1		10					"
		102			4/							"										"	D2		"					"
		103				4/						"										"	D3		"					"
		104					4/					"										"	D4		"					"
		105						4/				"										"	D5		"					"
		106							4/			"										"	D6		"					"
		107								4/		"										"	D7		"					"
		108									4/	"										"	D8		"					"
C_C		109										"	4/									"	CLK		15					"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2, X	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	Q8	Q7	Q6	Q5	Q4	Q3	Q2	Q1	V_{CC}		Min	Max	Min	Max	Min	Max	
C_O	3012	110	6V									GND										6V	Q1		20					pF
		111	"									"										"	Q2		"					"
		112	"									"										"	Q3		"					"
		113	"									"										"	Q4		"					"
		114	"									"										"	Q5		"					"
		115	"									"										"	Q6		"					"
		116	"									"										"	Q7		"					"
		117	"									"		4/	4/	4/	4/	4/	4/	4/	4/	"	Q8		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C =+125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	118	B	A	A	A	A	A	A	A	A	"	B	X	X	X	X	X	X	X	X	4.5V	All outputs 7/	8/	8/					
		119	"	A	B	A	A	B	A	A	A	"	A	H	H	H	H	H	H	H	H	"								
		120	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		121	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		122	"	"	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"									
		123	"	"	"	"	"	"	"	"	"	"	B	L	L	L	L	L	L	L	L	"									
		124	"	A	A	A	A	A	A	A	A	"	A	H	H	H	H	H	H	H	H	"								
																								Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
																								Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		125	GND	IN								GND	IN									4.5V	Q1	33		25		33		MHz
		126	"		IN							"	"									"	Q2	"		"		"		"
		127	"			IN						"	"									"	Q3	"		"		"		"
		128	"				IN					"	"									"	Q4	"		"		"		"
		129	"					IN				"	"									"	Q5	"		"		"		"
		130	"						IN			"	"									"	Q6	"		"		"		"
		131	"							IN		"	"									"	Q7	"		"		"		"
		132	"								IN	"	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	Q8	"		"		"		"
t_{PHL2}	3003 (Fig. 3)	133	"	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"									"	CLK to Q1		32		42		32	ns
		134	"	"	IN	4.5V	4.5V	"	"	"	"	"	"									"	CLK to Q2		"		"		"	"
		135	"	"	"	IN	4.5V	IN	"	"	"	"	"									"	CLK to Q3		"		"		"	"
		136	"	"	"	"	IN	4.5V	IN	"	"	"	"									"	CLK to Q4		"		"		"	"
		137	"	"	"	"	"	"	4.5V	IN	"	"	"									"	CLK to Q5		"		"		"	"
		138	"	"	"	"	"	"	"	4.5V	IN	"	"									"	CLK to Q6		"		"		"	"
		139	"	"	"	"	"	"	"	"	4.5V	"	"									"	CLK to Q7		"		"		"	"
		140	"	"	"	"	"	"	"	"	IN	"	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	CLK to Q8		"		"		"	"
t_{PLH2}		141	"	IN	GND	GND	GND	GND	GND	GND	GND	"	"									"	CLK to Q1		"		"		"	"
		142	"	GND	IN	GND	GND	GND	GND	GND	GND	"	"									"	CLK to Q2		"		"		"	"
		143	"	"	GND	IN	GND	"	"	"	"	"	"									"	CLK to Q3		"		"		"	"
		144	"	"	"	GND	IN	GND	"	"	"	"	"									"	CLK to Q4		"		"		"	"
		145	"	"	"	"	GND	IN	GND	"	"	"	"									"	CLK to Q5		"		"		"	"
		146	"	"	"	"	"	GND	IN	GND	"	"	"									"	CLK to Q6		"		"		"	"
		147	"	"	"	"	"	"	GND	IN	GND	"	"									"	CLK to Q7		"		"		"	"
		148	"	"	"	"	"	"	"	GND	IN	"	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	CLK to Q8		"		"		"	"
t_{PZH}		149	IN	4.5V								"										"	$\overline{OE}$ to Q1		26		35		26	"
		150	"		4.5V							"										"	$\overline{OE}$ to Q2		"		"		"	"
		151	"			4.5V						"										"	$\overline{OE}$ to Q3		"		"		"	"
		152	"				4.5V					"										"	$\overline{OE}$ to Q4		"		"		"	"
		153	"					4.5V				"										"	$\overline{OE}$ to Q5		"		"		"	"
		154	"						4.5V			"										"	$\overline{OE}$ to Q6		"		"		"	"
		155	"							4.5V		"										"	$\overline{OE}$ to Q7		"		"		"	"
		156	"								4.5V	"		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	$\overline{OE}$ to Q8		"		"		"	"
t_{PZL}		157	"	GND								"										"	$\overline{OE}$ to Q1		"		"		"	"
		158	"		GND							"										"	$\overline{OE}$ to Q2		"		"		"	"
		159	"			GND						"										"	$\overline{OE}$ to Q3		"		"		"	"
		160	"				GND					"										"	$\overline{OE}$ to Q4		"		"		"	"
		161	"					GND				"										"	$\overline{OE}$ to Q5		"		"		"	"
		162	"						GND			"										"	$\overline{OE}$ to Q6		"		"		"	"
		163	"							GND		"										"	$\overline{OE}$ to Q7		"		"		"	"
		164	"								GND	"		OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	$\overline{OE}$ to Q8		"		"		"	"

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TABLE III. Group A inspection for device type 04 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2, X	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	Q8	Q7	Q6	Q5	Q4	Q3	Q2	Q1	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PHZ}	3003 (Fig. 3)	165	IN	4.5V	4.5V							GND										4.5V	$\overline{OE}$ to Q1		26		35		26	ns
		166	"			4.5V						"								OUT	OUT	"	$\overline{OE}$ to Q2		"		"		"	"
		167	"				4.5V					"							OUT			"	$\overline{OE}$ to Q3		"		"		"	"
		168	"					4.5V				"						OUT				"	$\overline{OE}$ to Q4		"		"		"	"
		169	"						4.5V			"					OUT					"	$\overline{OE}$ to Q5		"		"		"	"
		170	"							4.5V		"				OUT						"	$\overline{OE}$ to Q6		"		"		"	"
		171	"								4.5V	"			OUT							"	$\overline{OE}$ to Q7		"		"		"	"
		172	"									"		OUT								"	$\overline{OE}$ to Q8		"		"		"	"
t_{PLZ}		173	"	GND								"										"	$\overline{OE}$ to Q1		"		"		"	"
		174	"		GND							"								OUT	OUT	"	$\overline{OE}$ to Q2		"		"		"	"
		175	"			GND						"							OUT			"	$\overline{OE}$ to Q3		"		"		"	"
		176	"				GND					"						OUT				"	$\overline{OE}$ to Q4		"		"		"	"
		177	"					GND				"					OUT					"	$\overline{OE}$ to Q5		"		"		"	"
		178	"						GND			"				OUT						"	$\overline{OE}$ to Q6		"		"		"	"
		179	"							GND		"			OUT							"	$\overline{OE}$ to Q7		"		"		"	"
		180	"								GND	"		OUT								"	$\overline{OE}$ to Q8		"		"		"	"
t_{THL}	3004	181	GND	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	GND	IN									4.5V	Q1		12		16		12	ns
		182	"	4.5V	IN	4.5V	"	"	"	"	"	"	"							OUT	OUT	"	Q2		"		"		"	"
		183	"	"	4.5V	IN	"	"	"	"	"	"	"						OUT			"	Q3		"		"		"	"
		184	"	"	"	4.5V	IN	"	"	"	"	"	"					OUT				"	Q4		"		"		"	"
		185	"	"	"	"	4.5V	IN	"	"	"	"	"				OUT					"	Q5		"		"		"	"
		186	"	"	"	"	"	4.5V	IN	"	"	"	"			OUT						"	Q6		"		"		"	"
		187	"	"	"	"	"	"	4.5V	IN	"	"	"		OUT							"	Q7		"		"		"	"
		188	"	"	"	"	"	"	"	4.5V	IN	"	"	OUT								"	Q8		"		"		"	"
t_{TLH}		189	"	IN	GND	GND	GND	GND	GND	GND	GND	"	"									"	Q1		"		"		"	"
		190	"	GND	IN	GND	"	"	"	"	"	"	"							OUT	OUT	"	Q2		"		"		"	"
		191	"	"	GND	IN	"	"	"	"	"	"	"						OUT			"	Q3		"		"		"	"
		192	"	"	"	GND	IN	"	"	"	"	"	"					OUT				"	Q4		"		"		"	"
		193	"	"	"	"	GND	IN	"	"	"	"	"				OUT					"	Q5		"		"		"	"
		194	"	"	"	"	"	GND	IN	"	"	"	"			OUT						"	Q6		"		"		"	"
		195	"	"	"	"	"	"	GND	IN	"	"	"		OUT							"	Q7		"		"		"	"
		196	"	"	"	"	"	"	"	GND	IN	"	"	OUT								"	Q8		"		"		"	"

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	$\overline{Q1}$	D1	D2	$\overline{Q2}$	$\overline{Q3}$	D3	D4	$\overline{Q4}$	GND	CLK	$\overline{Q5}$	D5	D6	$\overline{Q6}$	$\overline{Q7}$	D7	D8	$\overline{Q8}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA		1mA							1/										GND	$\overline{OE}$	2/	1.5					V
		2				1mA			1mA			"										"	D1		"					"
		3								1mA		"	1mA		1mA							"	D2		"					"
		4										"				1mA						"	D3		"					"
		5										"							1mA			"	D4		"					"
		6										"								1mA		"	CLK		"					"
		7										"										"	D5		"					"
		8										"										"	D6		"					"
		9										"										"	D7		"					"
		10										"										"	D8		"					"
V_{IC}(neg)		11	-1mA		-1mA							GND										1/	$\overline{OE}$		-1.5					"
		12				-1mA			-1mA			"										"	D1		"					"
		13								-1mA		"	-1mA									"	D2		"					"
		14										"			-1mA							"	D3		"					"
		15										"				-1mA						"	D4		"					"
		16										"										"	CLK		"					"
		17										"							-1mA			"	D5		"					"
		18										"								-1mA		"	D6		"					"
		19										"										"	D7		"					"
		20										"										"	D8		"					"
I_{CCH}	3005	21	GND		GND	GND			GND	GND		"	3/		GND	GND			GND	GND		6V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND		6V	6V			6V	6V		"	"		6V	6V			6V	6V		"	"		0.2		8.0			"
I_{CCZ}		23	6V		6V	6V			6V	6V		"	"		6V	6V			6V	6V		"	"		.1		4			"
V_{OH3}	3006	24	1.2V	-20µA	1.2V							"	"									"	O1	5.95		5.95		5.95		V
		25	"	"	"	1.2V	-20µA					"	"									"	O2	"		"		"		"
		26	"	"	"			-20µA	1.2V			"	"									"	O3	"		"		"		"
		27	"	"	"					1.2V	-20µA	"	"									"	O4	"		"		"		"
		28	"	"	"							"	"	-20µA	1.2V							"	O5	"		"		"		"
		29	"	"	"							"	"			1.2V	-20µA					"	O6	"		"		"		"
		30	"	"	"							"	"					-20µA	1.2V			"	O7	"		"		"		"
		31	"	"	"							"	"							1.2V	-20µA	"	O8	"		"		"		"
V_{OH5}		32	"	-5.2mA	1.2V							"	"									"	O1	5.48		5.2		5.48		"
		33	"	"	"	1.2V	-5.2mA					"	"									"	O2	"		"		"		"
		34	"	"	"			-5.2mA	1.2V			"	"									"	O3	"		"		"		"
		35	"	"	"					1.2V	-5.2mA	"	"									"	O4	"		"		"		"
		36	"	"	"							"	"	-5.2mA	1.2V							"	O5	"		"		"		"
		37	"	"	"							"	"			1.2V	-5.2mA					"	O6	"		"		"		"
		38	"	"	"							"	"					-5.2mA	1.2V			"	O7	"		"		"		"
		39	"	"	"							"	"							1.2V	-5.2mA	"	O8	"		"		"		"
V_{OL3}	3007	40	"	20µA	4.2V							"	"									"	O1		0.1		0.1		0.1	"
		41	"	"	"	4.2V	20µA					"	"									"	O2		"		"		"	"
		42	"	"	"			20µA	4.2V			"	"									"	O3		"		"		"	"
		43	"	"	"					4.2V	20µA	"	"									"	O4		"		"		"	"
		44	"	"	"							"	"	20µA	4.2V							"	O5		"		"		"	"
		45	"	"	"							"	"			4.2V	20µA					"	O6		"		"		"	"
		46	"	"	"							"	"					20µA	4.2V			"	O7		"		"		"	"
		47	"	"	"							"	"							4.2V	20µA	"	O8		"		"		"	"
V_{OL5}		48	"	5.2mA	4.2V							"	"									"	O1		0.26		0.4		0.26	"
		49	"	"	"	4.2V	5.2mA					"	"									"	O2		"		"		"	"
		50	"	"	"			5.2mA	4.2V			"	"									"	O3		"		"		"	"
		51	"	"	"					4.2V	5.2mA	"	"									"	O4		"		"		"	"
		52	"	"	"							"	"	5.2mA	4.2V							"	O5		"		"		"	"
		53	"	"	"							"	"			4.2V	5.2mA					"	O6		"		"		"	"
		54	"	"	"							"	"					5.2mA	4.2V			"	O7		"		"		"	"
		55	"	"	"							"	"							4.2V	5.2mA	"	O8		"		"		"	"

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TABLE III. Group A inspection for device type 05 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	$\overline{Q1}$	D1	D2	$\overline{Q2}$	$\overline{Q3}$	D3	D4	$\overline{Q4}$	GND	CLK	$\overline{Q5}$	D5	D6	$\overline{Q6}$	$\overline{Q7}$	D7	D8	$\overline{Q8}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	3/	GND	GND	GND	GND	GND	GND	GND		4V	$\overline{Q1}$	-10	-135	-10	-135	-10	-135	mA
		57	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q2}$	"	"	"	"	"	"	"
		58	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q3}$	"	"	"	"	"	"	"
		59	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q4}$	"	"	"	"	"	"	"
		60	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q5}$	"	"	"	"	"	"	"
		61	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q6}$	"	"	"	"	"	"	"
		62	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q7}$	"	"	"	"	"	"	"
		63	"		"	"			"	"		"	"		"	"			"	"		"	$\overline{Q8}$	"	"	"	"	"	"	"
I_{IH}	3010	64	6V		"	"			"	"		"	GND		"	"			"	"		6V	$\overline{OE}$		.05		0.1			µA
		65	GND		6V	6V			"	"		"	"		"	"			"	"		"	D1		"		"			"
		66	"		GND	GND			"	"		"	"		"	"			"	"		"	D2		"		"			"
		67	"		"	"			6V	"		"	"		"	"			"	"		"	D3		"		"			"
		68	"		"	"			GND	6V		"	"		"	"			"	"		"	D4		"		"			"
		69	"		"	"			"	GND		"	6V		"	"			"	"		"	CLK		"		"			"
		70	"		"	"			"	"		"	GND		6V	"			"	"		"	D5		"		"			"
		71	"		"	"			"	"		"	"		GND	6V			"	"		"	D6		"		"			"
		72	"		"	"			"	"		"	"		"	GND			6V	"		"	D7		"		"			"
		73	"		"	"			"	"		"	"		"	"			GND	6V		"	D8		"		"			"
I_{IL}	3009	74	"		6V	6V			6V	6V		"	6V		6V	6V			6V	"		"	$\overline{OE}$		-.05		-0.1			"
		75	6V		GND	GND			"	"		"	"		"	"			"	"		"	D1		"		"			"
		76	"		6V	GND			GND	"		"	"		"	"			"	"		"	D2		"		"			"
		77	"		"	6V			"	GND		"	"		"	"			"	"		"	D3		"		"			"
		78	"		"	"			6V	6V		"	"		"	"			"	"		"	D4		"		"			"
		79	"		"	"			"	"		"	GND		"	"			"	"		"	CLK		"		"			"
		80	"		"	"			"	"		"	6V		GND	"			"	"		"	D5		"		"			"
		81	"		"	"			"	"		"	"		6V	GND			"	"		"	D6		"		"			"
		82	"		"	"			"	"		"	"		"	6V			GND	"		"	D7		"		"			"
		83	"		"	"			"	"		"	"		"	"			6V	GND		"	D8		"		"			"
I_{OZH}		84	4.2V	6.0V	1.2V		6.0V		1.2V		6.0V	"	3/		1.2V		6.0V		1.2V		6.0V	"	$\overline{Q1}$		0.2		2.0			"
		85	"		"	1.2V		6.0V	"	1.2V		"	"		"	1.2V		6.0V	"		"	"	$\overline{Q2}$		"		"			"
		86	"		"	"			"	"		"	"		"	"			"		"	"	$\overline{Q3}$		"		"			"
		87	"		"	"			"	1.2V		"	"		"	"			"		"	"	$\overline{Q4}$		"		"			"
		88	"		"	"			"	"		"	"		"	"			"		"	"	$\overline{Q5}$		"		"			"
		89	"		"	"			"	"		"	"		"	"			"		"	"	$\overline{Q6}$		"		"			"
		90	"		"	"			"	"		"	"		"	"			"		"	"	$\overline{Q7}$		"		"			"
		91	"		"	"			"	"		"	"		"	"			"		"	"	$\overline{Q8}$		"		"			"
I_{OZL}		92	"		"				"			"	"		"				"			"	$\overline{Q1}$		-0.2		-2.0			"
		93	"		"	4.2V			"			"	"		"				"			"	$\overline{Q2}$		"		"			"
		94	"		"	"			"			"	"		"				"			"	$\overline{Q3}$		"		"			"
		95	"		"	"			"	4.2V		"	"		"				"			"	$\overline{Q4}$		"		"			"
		96	"		"	"			"	"		"	"		"				"			"	$\overline{Q5}$		"		"			"
		97	"		"	"			"	"		"	"		"				"			"	$\overline{Q6}$		"		"			"
		98	"		"	"			"	"		"	"		"				"			"	$\overline{Q7}$		"		"			"
		99	"		"	"			"	"		"	"		"				"			"	$\overline{Q8}$		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	100	4/									GND										GND	$\overline{OE}$		15					pF
C_I		101			4/							"										"	D1		10					"
		102				4/						"										"	D2		"					"
		103							4/			"										"	D3		"					"
		104								4/		"										"	D4		"					"
C_C		105										"	4/									"	CLK		15					"
C_I		106										"			4/							"	D5		10					"
		107										"				4/						"	D6		"					"
		108										"							4/			"	D7		"					"
		109										"								4/			"	D8		"					"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	$\overline{Q1}$	D1	D2	$\overline{Q2}$	$\overline{Q3}$	D3	D4	$\overline{Q4}$	GND	CLK	$\overline{Q5}$	D5	D6	$\overline{Q6}$	$\overline{Q7}$	D7	D8	$\overline{Q8}$	V_{CC}		Min	Max	Min	Max	Min	Max	
C_O	3012	110 111 112 113 114 115 116 117	6V “ “ “ “ “ “ “	4/			4/	4/			4/	GND “ “ “ “ “ “ “		4/			4/	4/			4/	6V “ “ “ “ “ “ “	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8		20 “ “ “ “ “ “ “					pF “ “ “ “ “ “ “
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	118 119 120 121 122 123 124 125 126 127 128 129 130	B “ “ “ “ “ “ “ “ “ “ “ “	H “ L “ H “ L “ H	B A “ B “ A “ B	A B “ A “ B “ A	L “ H “ L “ H “ L	H “ L “ H “ L “ H	B A “ B “ A “ B	A B “ A “ B “ A	L “ H “ L “ H “ L	GND “ “ “ “ “ “ “ “	A B B A B A B B A	H “ L “ H “ L “ H	B A “ B “ A “ B	A B “ A “ B “ A	L “ H “ L “ H “ L	H “ L “ H “ L “ H	B A “ B “ A “ B	A B “ A “ B “ A	L “ H “ L “ H “ L	4.5V “ “ “ “ “ “ “ “	All outputs 7/ 8/						
																								Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
																								Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		131 132 133 134 135 136 137 138	GND “ “ “ “ “ “ “	OUT	IN	IN	OUT	OUT	IN	IN	OUT	GND “ “ “ “ “ “ “	IN “ “ “ “ “ “ “	OUT	IN	IN	OUT	OUT	IN	IN	OUT	4.5V “ “ “ “ “ “ “	Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8	33 “ “ “ “ “ “ “		25 “ “ “ “ “ “ “		33 “ “ “ “ “ “ “		MHz “ “ “ “ “ “ “
t_{PHL2}	3003 (Fig.3)	139 140 141 142 143 144 145 146	“ “ “ “ “ “ “ “	OUT	IN GND “ “ “ “ “ “	GND IN GND “ “ “ “ “	OUT	OUT	GND GND IN GND “ “ “ “	GND “ IN GND “ “ “ “	OUT	“ “ “ “ “ “ “ “	“ “ “ “ “ “ “ “	OUT	GND “ “ “ IN GND “ “	GND “ “ “ IN GND GND “	OUT	OUT	GND “ “ “ “ “ IN GND	GND “ “ “ “ “ IN “	OUT	“ “ “ “ “ “ “ “	CLK to Q1 CLK to Q2 CLK to Q3 CLK to Q4 CLK to Q5 CLK to Q6 CLK to Q7 CLK to Q8		32 “ “ “ “ “ “ “		42 “ “ “ “ “ “ “		32 “ “ “ “ “ “ “	ns “ “ “ “ “ “ “
t_{PLH2}		147 148 149 150 151 152 153 154	“ “ “ “ “ “ “ “	OUT	IN 4.5V “ “ “ “ “ “	4.5V IN 4.5V “ “ “ “ “	OUT	OUT	4.5V 4.5V IN 4.5V “ “ “ “	4.5V “ IN 4.5V “ “ “ “	OUT	“ “ “ “ “ “ “ “	“ “ “ “ “ “ “ “	OUT	4.5V “ “ “ IN 4.5V “ “	4.5V “ “ “ IN 4.5V 4.5V “	OUT	OUT	4.5V “ “ “ “ “ IN 4.5V	4.5V “ “ “ “ “ IN “	OUT	“ “ “ “ “ “ “ “	CLK to Q1 CLK to Q2 CLK to Q3 CLK to Q4 CLK to Q5 CLK to Q6 CLK to Q7 CLK to Q8		“ “ “ “ “ “ “ “		“ “ “ “ “ “ “ “		“ “ “ “ “ “ “ “	“ “ “ “ “ “ “ “
t_{PZH}		155 156 157 158 159 160 161 162	IN “ “ “ “ “ “ “	OUT	GND	GND	OUT	OUT	GND	GND	OUT	“ “ “ “ “ “ “ “		OUT	GND	GND	OUT	OUT	GND	GND	OUT	“ “ “ “ “ “ “ “	OE Q1 OE Q2 OE Q3 OE Q4 OE Q5 OE Q6 OE Q7 OE Q8		31 “ “ “ “ “ “ “		41 “ “ “ “ “ “ “		31 “ “ “ “ “ “ “	“ “ “ “ “ “ “ “

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TABLE III. Group A inspection for device type 05 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{OE}$	$\overline{Q1}$	D1	D2	$\overline{Q2}$	$\overline{Q3}$	D3	D4	$\overline{Q4}$	GND	CLK	$\overline{Q5}$	D5	D6	$\overline{Q6}$	$\overline{Q7}$	D7	D8	$\overline{Q8}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZL}	3003 (Fig. 3)	163	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	GND		OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	4.5V	$\overline{OE}$ $\overline{Q1}$		31		41		31	ns
		164	"									"										"	$\overline{OE}$ $\overline{Q2}$		"		"		"	"
		165	"									"										"	$\overline{OE}$ $\overline{Q3}$		"		"		"	"
		166	"									"										"	$\overline{OE}$ $\overline{Q4}$		"		"		"	"
		167	"									"										"	$\overline{OE}$ $\overline{Q5}$		"		"		"	"
		168	"									"										"	$\overline{OE}$ $\overline{Q6}$		"		"		"	"
		169	"									"										"	$\overline{OE}$ $\overline{Q7}$		"		"		"	"
		170	"									"										"	$\overline{OE}$ $\overline{Q8}$		"		"		"	"
t_{PHZ}		171	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"		OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ $\overline{Q1}$		26		35		26	"
		172	"									"										"	$\overline{OE}$ $\overline{Q2}$		"		"		"	"
		173	"									"										"	$\overline{OE}$ $\overline{Q3}$		"		"		"	"
		174	"									"										"	$\overline{OE}$ $\overline{Q4}$		"		"		"	"
		175	"									"										"	$\overline{OE}$ $\overline{Q5}$		"		"		"	"
		176	"									"										"	$\overline{OE}$ $\overline{Q6}$		"		"		"	"
		177	"									"										"	$\overline{OE}$ $\overline{Q7}$		"		"		"	"
		178	"									"										"	$\overline{OE}$ $\overline{Q8}$		"		"		"	"
t_{PLZ}		179	IN	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	GND		OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{OE}$ $\overline{Q1}$		26		35		26	ns
		180	"									"										"	$\overline{OE}$ $\overline{Q2}$		"		"		"	"
		181	"									"										"	$\overline{OE}$ $\overline{Q3}$		"		"		"	"
		182	"									"										"	$\overline{OE}$ $\overline{Q4}$		"		"		"	"
		183	"									"										"	$\overline{OE}$ $\overline{Q5}$		"		"		"	"
		184	"									"										"	$\overline{OE}$ $\overline{Q6}$		"		"		"	"
		185	"									"										"	$\overline{OE}$ $\overline{Q7}$		"		"		"	"
		186	"									"										"	$\overline{OE}$ $\overline{Q8}$		"		"		"	"
t_{THL}	3004	187	GND	OUT	IN	GND	OUT	OUT	GND	GND	OUT	"	IN	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{Q1}$		12		16		12	ns
		188	"		GND	IN			GND	"		"	"		"	"			"	"		"	$\overline{Q2}$		"		"		"	"
		189	"		"	GND			IN	"		"	"		"	"			"	"		"	$\overline{Q3}$		"		"		"	"
		190	"		"	"			GND	IN		"	"		"	"			"	"		"	$\overline{Q4}$		"		"		"	"
		191	"		"	"			"	GND		"	"		IN	"			"	"		"	$\overline{Q5}$		"		"		"	"
		192	"		"	"			"	"		"	"		GND	IN			"	"		"	$\overline{Q6}$		"		"		"	"
		193	"		"	"			"	"		"	"		"	GND			IN	"		"	$\overline{Q7}$		"		"		"	"
		194	"		"	"			"	"		"	"		"	GND			GND	IN		"	$\overline{Q8}$		"		"		"	"
t_{TLH}		195	"	OUT	IN	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	"	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{Q1}$		"		"		"	"
		196	"		4.5V	IN			4.5V	"		"	"		"	"			"	"		"	$\overline{Q2}$		"		"		"	"
		197	"		"	4.5V			IN	"		"	"		"	"			"	"		"	$\overline{Q3}$		"		"		"	"
		198	"		"	"			4.5V	IN		"	"		"	"			"	"		"	$\overline{Q4}$		"		"		"	"
		199	"		"	"			"	4.5V		"	"		IN	"			"	"		"	$\overline{Q5}$		"		"		"	"
		200	"		"	"			"	"		"	"		4.5V	IN			"	"		"	$\overline{Q6}$		"		"		"	"
		201	"		"	"			"	"		"	"		"	4.5V			IN	"		"	$\overline{Q7}$		"		"		"	"
		202	"		"	"			"	"		"	"		"	4.5V			4.5V	IN		"	$\overline{Q8}$		"		"		"	"

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TABLE III. Group A inspection for device type 06.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	$\overline{Q8}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1mA	1/	1mA									GND	$\overline{OE}$	2/	1.5					V
		2										"										"	D1		"					"
		3										"										"	D2		"					"
		4										"										"	D3		"					"
		5										"										"	D4		"					"
		6										"										"	D5		"					"
		7										"										"	D6		"					"
		8										"										"	D7		"					"
		9										"										"	D8		"					"
		10										"	1mA									"	CLK		"					"
V_{IC}(neg)		11	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA	-1mA	GND										1/	$\overline{OE}$		-1.5					"
		12										"										"	D1		"					"
		13										"										"	D2		"					"
		14										"										"	D3		"					"
		15										"										"	D4		"					"
		16										"										"	D5		"					"
		17										"										"	D6		"					"
		18										"										"	D7		"					"
		19										"										"	D8		"					"
		20										"	-1mA									"	CLK		"					"
I_{CCH}	3005	21	GND	GND	GND	GND	GND	GND	GND	GND	GND	"	3/									6V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND	6V	6V	6V	6V	6V	6V	6V	6V	"	"									"	"		0.2		8.0			"
I_{CCZ}		23	6V	6V	6V	6V	6V	6V	6V	6V	6V	"	"									"	"		.1		4			"
V_{OH3}	3006	24	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	"	"								-20µA	"	$\overline{Q1}$	5.95		5.95		5.95		V
		25	"									"	"									"	$\overline{Q2}$	"		"		"		"
		26	"									"	"							-20µA		"	$\overline{Q3}$	"		"		"		"
		27	"									"	"									"	$\overline{Q4}$	"		"		"		"
		28	"									"	"					-20µA				"	$\overline{Q5}$	"		"		"		"
		29	"									"	"			-20µA						"	$\overline{Q6}$	"		"		"		"
		30	"									"	"		-20µA							"	$\overline{Q7}$	"		"		"		"
		31	"									"	"	-20µA								"	Q8	"		"		"		"
V_{OH5}		32	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	"	"								-5.2mA	"	$\overline{Q1}$	5.48		5.2		5.48		"
		33	"									"	"							-5.2mA		"	$\overline{Q2}$	"		"		"		"
		34	"									"	"						-5.2mA			"	$\overline{Q3}$	"		"		"		"
		35	"									"	"					-5.2mA				"	$\overline{Q4}$	"		"		"		"
		36	"									"	"				-5.2mA					"	$\overline{Q5}$	"		"		"		"
		37	"									"	"			-5.2mA						"	$\overline{Q6}$	"		"		"		"
		38	"									"	"		-5.2mA							"	$\overline{Q7}$	"		"		"		"
		39	"									"	"	-5.2mA								"	Q8	"		"		"		"
V_{OL3}	3007	40	1.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	"	"								20µA	"	$\overline{Q1}$		0.1		0.1		0.1	"
		41	"									"	"							20µA		"	$\overline{Q2}$		"		"		"	"
		42	"									"	"						20µA			"	$\overline{Q3}$		"		"		"	"
		43	"									"	"					20µA				"	$\overline{Q4}$		"		"		"	"
		44	"									"	"				20µA					"	$\overline{Q5}$		"		"		"	"
		45	"									"	"			20µA						"	$\overline{Q6}$		"		"		"	"
		46	"									"	"		20µA							"	$\overline{Q7}$		"		"		"	"
		47	"									"	"	20µA								"	Q8		"		"		"	"
V_{OL5}		48	1.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	"	"								5.2mA	"	$\overline{Q1}$		0.26		0.4		0.26	"
		49	"									"	"							5.2mA		"	$\overline{Q2}$		"		"		"	"
		50	"									"	"						5.2mA			"	$\overline{Q3}$		"		"		"	"
		51	"									"	"					5.2mA				"	$\overline{Q4}$		"		"		"	"
		52	"									"	"				5.2mA					"	$\overline{Q5}$		"		"		"	"
		53	"									"	"			5.2mA						"	$\overline{Q6}$		"		"		"	"
		54	"									"	"		5.2mA							"	$\overline{Q7}$		"		"		"	"
		55	"									"	"	5.2mA								"	Q8		"		"		"	"

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	$\overline{Q8}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	3/	GND	GND	GND	GND	GND	GND	GND	GND	4V	$\overline{Q1}$	-10	-135	-10	-135	-10	-135	mA
		57	"									"	"									"	$\overline{Q2}$	"	"	"	"	"	"	"
		58	"									"	"									"	$\overline{Q3}$	"	"	"	"	"	"	"
		59	"									"	"									"	$\overline{Q4}$	"	"	"	"	"	"	"
		60	"									"	"									"	$\overline{Q5}$	"	"	"	"	"	"	"
		61	"									"	"									"	$\overline{Q6}$	"	"	"	"	"	"	"
		62	"									"	"									"	$\overline{Q7}$	"	"	"	"	"	"	"
		63	"									"	"									"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	64	6V	GND	GND	GND	GND	GND	GND	GND	GND	"										6V	$\overline{OE}$		.05		0.1			µA
		65	GND	6V	GND	"	"	"	"	"	"	"										"	D1		"		"			"
		66	"	GND	6V	"	"	"	"	"	"	"										"	D2		"		"			"
		67	"	"	GND	6V	"	"	"	"	"	"										"	D3		"		"			"
		68	"	"	"	GND	6V	"	"	"	"	"										"	D4		"		"			"
		69	"	"	"	"	GND	6V	"	"	"	"										"	D5		"		"			"
		70	"	"	"	"	"	GND	6V	"	"	"										"	D6		"		"			"
		71	"	"	"	"	"	"	GND	6V	"	"										"	D7		"		"			"
		72	"	"	"	"	"	"	"	GND	6V	"		GND								"	D8		"		"			"
		73	"	"	"	"	"	"	"	GND	GND	"		6V								"	CLK		"		"			"
I_{IL}	3009	74	"	6V	6V	6V	6V	6V	6V	6V	6V	"	"									"	$\overline{OE}$		-.05		-0.1			"
		75	6V	GND	6V	"	"	"	"	"	"	"	"									"	D1		"		"			"
		76	"	"	GND	6V	"	"	"	"	"	"	"									"	D2		"		"			"
		77	"	"	"	"	GND	"	"	"	"	"	"									"	D3		"		"			"
		78	"	"	"	GND	6V	GND	"	"	"	"	"									"	D4		"		"			"
		79	"	"	"	"	"	6V	GND	"	"	"	"									"	D5		"		"			"
		80	"	"	"	"	"	"	6V	GND	"	"	"									"	D6		"		"			"
		81	"	"	"	"	"	"	"	6V	GND	"	"									"	D7		"		"			"
		82	"	"	"	"	"	"	"	"	6V	"	"									"	D8		"		"			"
		83	"	"	"	"	"	"	"	"	"	"	GND									"	CLK		"		"			"
I_{OZH}		84	4.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	1.2V	"	3/	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	6.0V	"	$\overline{Q1}$		0.2		2.0			"
		85	"									"	"									"	$\overline{Q2}$		"		"			"
		86	"									"	"									"	$\overline{Q3}$		"		"			"
		87	"									"	"									"	$\overline{Q4}$		"		"			"
		88	"									"	"									"	$\overline{Q5}$		"		"			"
		89	"									"	"									"	$\overline{Q6}$		"		"			"
		90	"									"	"									"	$\overline{Q7}$		"		"			"
		91	"									"	"									"	Q8		"		"			"
I_{OZL}		92	"	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	4.2V	"	"	6.0V	GND	GND	GND	GND	GND	GND	GND	"	$\overline{Q1}$		-0.2		-2.0			"
		93	"									"	"									"	$\overline{Q2}$		"		"			"
		94	"									"	"									"	$\overline{Q3}$		"		"			"
		95	"									"	"									"	$\overline{Q4}$		"		"			"
		96	"									"	"									"	$\overline{Q5}$		"		"			"
		97	"									"	"									"	$\overline{Q6}$		"		"			"
		98	"									"	"									"	$\overline{Q7}$		"		"			"
		99	"									"	"									"	Q8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	100	4/									GND										GND	$\overline{OE}$		15					pF
C_I		101		4/								"										"	D1		10					"
		102			4/							"										"	D2		"					"
		103				4/						"										"	D3		"					"
		104					4/					"										"	D4		"					"
		105						4/				"										"	D5		"					"
		106							4/			"										"	D6		"					"
		107								4/		"										"	D7		"					"
		108									4/	"										"	D8		"					"
C_C		109										"	4/									"	CLK		15					"

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	$\overline{Q8}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	V_{CC}		Min	Max	Min	Max	Min	Max	
C_O	3012	110	6V									GND		4/	4/							6V	$\overline{Q8}$		20				pF
		111	"									"				4/						"	$\overline{Q7}$		"				"
		112	"									"					4/					"	$\overline{Q6}$		"				"
		113	"									"						4/				"	$\overline{Q5}$		"				"
		114	"									"							4/			"	$\overline{Q4}$		"				"
		115	"									"								4/		"	$\overline{Q3}$		"				"
		116	"									"									4/	"	$\overline{Q2}$		"				"
		117	"									"										"	$\overline{Q1}$		"				"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	118	B	B	B	B	B	B	B	B	B	"	B	X	X	X	X	X	X	X	X	4.5V	All outputs 7/	8/	8/				
		119	"	B	A	A	A	A	A	A	A	"	A	H	H	H	H	H	H	H	H	"							
		120	"	A	"	"	"	"	"	"	"	"	A	"	"	"	"	"	"	"	"	"								
		121	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"								
		122	"	"	B	B	B	B	B	B	B	"	A	L	L	L	L	L	L	L	L	"								
		123	"	"	B	B	B	B	B	B	B	"	B	L	L	L	L	L	L	L	L	"								
		124	"	"	B	B	B	B	B	B	B	"	A	H	H	H	H	H	H	H	H	"								
																								Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
																								Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		125	GND	IN	IN	IN	IN	IN	IN	IN	IN	GND	IN								OUT	4.5V	$\overline{Q1}$	33		25		33		MHz
		126	"									"	"									"	$\overline{Q2}$	"		"		"		"
		127	"									"	"						OUT			"	$\overline{Q3}$	"		"		"		"
		128	"									"	"					OUT		OUT		"	$\overline{Q4}$	"		"		"		"
		129	"									"	"				OUT					"	$\overline{Q5}$	"		"		"		"
		130	"									"	"			OUT						"	$\overline{Q6}$	"		"		"		"
		131	"									"	"		OUT							"	$\overline{Q7}$	"		"		"		"
		132	"									"	"	OUT								"	$\overline{Q8}$	"		"		"		"
t_{PHL2}	3003 (Fig. 3)	133	"	IN	GND	GND	GND	GND	GND	GND	GND	"	"								OUT	"	CLK to $\overline{Q1}$		31		41		31	ns
		134	"	GND	IN	GND	"	"	"	"	"	"	"									"	CLK to $\overline{Q2}$		"		"		"	"
		135	"	"	GND	IN	IN	"	"	"	"	"	"									"	CLK to $\overline{Q3}$		"		"		"	"
		136	"	"	"	GND	GND	IN	"	"	"	"	"									"	CLK to $\overline{Q4}$		"		"		"	"
		137	"	"	"	"	"	GND	IN	"	"	"	"									"	CLK to $\overline{Q5}$		"		"		"	"
		138	"	"	"	"	"	"	GND	IN	"	"	"									"	CLK to $\overline{Q6}$		"		"		"	"
		139	"	"	"	"	"	"	"	GND	IN	"	"									"	CLK to $\overline{Q7}$		"		"		"	"
		140	"	"	"	"	"	"	"	"	IN	"	"	OUT								"	CLK to $\overline{Q8}$		"		"		"	"
t_{PLH2}		141	"	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"								OUT	"	CLK to $\overline{Q1}$		"		"		"	"
		142	"	4.5V	IN	4.5V	"	"	"	"	"	"	"									"	CLK to $\overline{Q2}$		"		"		"	"
		143	"	"	4.5V	IN	IN	"	"	"	"	"	"									"	CLK to $\overline{Q3}$		"		"		"	"
		144	"	"	"	4.5V	"	IN	"	"	"	"	"									"	CLK to $\overline{Q4}$		"		"		"	"
		145	"	"	"	"	"	"	IN	"	"	"	"									"	CLK to $\overline{Q5}$		"		"		"	"
		146	"	"	"	"	"	"	"	IN	"	"	"									"	CLK to $\overline{Q6}$		"		"		"	"
		147	"	"	"	"	"	"	"	4.5V	IN	"	"									"	CLK to $\overline{Q7}$		"		"		"	"
		148	"	"	"	"	"	"	"	4.5V	IN	"	"	OUT								"	CLK to $\overline{Q8}$		"		"		"	"
t_{PZH}		149	IN	GND								"									OUT	"	$\overline{OE}$ to Q1		26		35		26	"
		150	"		GND							"										"	$\overline{OE}$ to $\overline{Q2}$		"		"		"	"
		151	"			GND						"										"	$\overline{OE}$ to $\overline{Q3}$		"		"		"	"
		152	"				GND					"										"	OE to Q4		"		"		"	"
		153	"					GND				"										"	$\overline{OE}$ to $\overline{Q5}$		"		"		"	"
		154	"						GND			"										"	$\overline{OE}$ to $\overline{Q6}$		"		"		"	"
		155	"							GND		"										"	$\overline{OE}$ to $\overline{Q7}$		"		"		"	"
		156	"								GND	"		OUT								"	OE to Q8		"		"		"	"

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TABLE III. Group A inspection for device type 06 – Continued.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{OE}$	D1	D2	D3	D4	D5	D6	D7	D8	GND	CLK	$\overline{Q8}$	$\overline{Q7}$	$\overline{Q6}$	$\overline{Q5}$	$\overline{Q4}$	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZL}	3003 (Fig. 3)	157	IN	4.5V								GND									OUT	4.5V	$\overline{OE}$ to $\overline{Q1}$		26		35		26	ns
		158	"		4.5V							"								OUT		"	$\overline{OE}$ to $\overline{Q2}$		"		"		"	"
		159	"			4.5V						"							OUT			"	$\overline{OE}$ to $\overline{Q3}$		"		"		"	"
		160	"				4.5V					"						OUT				"	$\overline{OE}$ to $\overline{Q4}$		"		"		"	"
		161	"					4.5V				"					OUT					"	$\overline{OE}$ to $\overline{Q5}$		"		"		"	"
		162	"						4.5V			"				OUT						"	$\overline{OE}$ to $\overline{Q6}$		"		"		"	"
		163	"							4.5V		"			OUT							"	$\overline{OE}$ to $\overline{Q7}$		"		"		"	"
		164	"								4.5V	"		OUT								"	$\overline{OE}$ to $\overline{Q8}$		"		"		"	"
t_{PHZ}		165	"	GND								"									OUT	"	$\overline{OE}$ to $\overline{Q1}$		"		"		"	"
		166	"		GND							"								OUT		"	$\overline{OE}$ to $\overline{Q2}$		"		"		"	"
		167	"			GND						"							OUT			"	$\overline{OE}$ to $\overline{Q3}$		"		"		"	"
		168	"				GND					"						OUT				"	$\overline{OE}$ to $\overline{Q4}$		"		"		"	"
		169	"					GND				"					OUT					"	$\overline{OE}$ to $\overline{Q5}$		"		"		"	"
		170	"						GND			"				OUT						"	$\overline{OE}$ to $\overline{Q6}$		"		"		"	"
		171	"							GND		"			OUT							"	$\overline{OE}$ to $\overline{Q7}$		"		"		"	"
		172	"								GND	"		OUT								"	$\overline{OE}$ to $\overline{Q8}$		"		"		"	"
t_{PLZ}		173	"	4.5V								"									OUT	"	$\overline{OE}$ to $\overline{Q1}$		"		"		"	"
		174	"		4.5V							"								OUT		"	$\overline{OE}$ to $\overline{Q2}$		"		"		"	"
		175	"			4.5V						"							OUT			"	$\overline{OE}$ to $\overline{Q3}$		"		"		"	"
		176	"				4.5V					"						OUT				"	$\overline{OE}$ to $\overline{Q4}$		"		"		"	"
		177	"					4.5V				"					OUT					"	$\overline{OE}$ to $\overline{Q5}$		"		"		"	"
		178	"						4.5V			"				OUT						"	$\overline{OE}$ to $\overline{Q6}$		"		"		"	"
		179	"							4.5V		"			OUT							"	$\overline{OE}$ to $\overline{Q7}$		"		"		"	"
		180	"								4.5V	"		OUT								"	$\overline{OE}$ to $\overline{Q8}$		"		"		"	"
t_{THL}	3004	181	GND	IN	GND	GND	GND	GND	GND	GND	GND	GND	IN								OUT	4.5V	$\overline{Q1}$		12		16		12	ns
		182	"	GND	IN	GND	"	"	"	"	"	"	"							OUT		"	$\overline{Q2}$		"		"		"	"
		183	"	"	GND	IN	IN	"	"	"	"	"	"						OUT			"	$\overline{Q3}$		"		"		"	"
		184	"	"	"	GND	GND	IN	"	"	"	"	"					OUT				"	$\overline{Q4}$		"		"		"	"
		185	"	"	"	"	"	GND	"	"	"	"	"				OUT					"	$\overline{Q5}$		"		"		"	"
		186	"	"	"	"	"	"	IN	"	"	"	"			OUT						"	$\overline{Q6}$		"		"		"	"
		187	"	"	"	"	"	"	GND	IN	"	"	"		OUT							"	$\overline{Q7}$		"		"		"	"
		188	"	"	"	"	"	"	"	GND	IN	"	"	OUT								"	Q8		"		"		"	"
t_{TLH}		189	"	IN	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	4.5V	"	"								OUT	"	$\overline{Q1}$		"		"		"	"
		190	"	4.5V	IN	4.5V	"	"	"	"	"	"	"							OUT		"	$\overline{Q2}$		"		"		"	"
		191	"	"	4.5V	IN	"	"	"	"	"	"	"						OUT			"	$\overline{Q3}$		"		"		"	"
		192	"	"	"	4.5V	IN	"	"	"	"	"	"					OUT				"	$\overline{Q4}$		"		"		"	"
		193	"	"	"	"	4.5V	IN	"	"	"	"	"				OUT					"	$\overline{Q5}$		"		"		"	"
		194	"	"	"	"	"	4.5V	IN	"	"	"	"			OUT						"	$\overline{Q6}$		"		"		"	"
		195	"	"	"	"	"	"	4.5V	IN	"	"	"		OUT							"	$\overline{Q7}$		"		"		"	"
		196	"	"	"	"	"	"	4.5V	4.5V	IN	"	"	OUT								"	Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 52.

Symbol	MIL-STD-883 method	Cases R,S,2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
V_{IC}(pos)		1	1mA		1mA							1/										GND	$\overline{OE}$	2/	1.5					V
		2				1mA			1mA			"										"	D1		"					"
		3								1mA		"										"	D2		"					"
		4										"	1mA									"	D3		"					"
		5										"										"	D4		"					"
		6										"										"	CLK		"					"
		7										"			1mA							"	D5		"					"
		8										"				1mA						"	D6		"					"
		9										"							1mA			"	D7		"					"
		10										"								1mA			"	D8		"					"
V_{IC}(neg)		11	-1mA									GND										1/	$\overline{OE}$		-1.5					"
		12			-1mA							"										"	D1		"					"
		13				-1mA						"										"	D2		"					"
		14							-1mA			"										"	D3		"					"
		15								-1mA		"										"	D4		"					"
		16										"	-1mA									"	CLK		"					"
		17										"			-1mA							"	D5		"					"
		18										"				-1mA						"	D6		"					"
		19										"							-1mA			"	D7		"					"
		20										"								-1mA			"	D8		"					"
I_{CCH}	3005	21	GND		5.5V	5.5V			5.5V	5.5V		"	3/		5.5V	5.5V			5.5V	5.5V		5.5V	V_{CC}		0.2		8.0			µA
I_{CCL}		22	GND		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	"		0.2		8.0			"
I_{CCZ}		23	5.5V		GND	GND			GND	GND		"	"		GND	GND			GND	GND		"	"		0.1		4.0			"
V_{OH6}	3006	24	0.8V	-20µA	2.0V							"	"									4.5V	Q1	4.4		4.4		4.4		V
		25	"	"		2.0V	-20µA					"	"									"	Q2	"		"		"		"
		26	"	"				-20µA	2.0V			"	"									"	Q3	"		"		"		"
		27	"	"						2.0V	-20µA	"	"									"	Q4	"		"		"		"
		28	"	"								"	"	-20µA	2.0V							"	Q5	"		"		"		"
		29	"	"								"	"			2.0V	-20µA					"	Q6	"		"		"		"
		30	"	"								"	"					-20µA	2.0V			"	Q7	"		"		"		"
		31	"	"								"	"							2.0V	-20µA	"	Q8	"		"		"		"
V_{OH7}		32	"	-6.0mA	2.0V							"	"									5.5V	Q1	3.98		3.7		3.98		"
		33	"	"		2.0V	-6.0mA					"	"									"	Q2	"		"		"		"
		34	"	"				-6.0mA	2.0V			"	"									"	Q3	"		"		"		"
		35	"	"						2.0V	-6.0mA	"	"									"	Q4	"		"		"		"
		36	"	"								"	"	-6.0mA	2.0V							"	Q5	"		"		"		"
		37	"	"								"	"			2.0V	-6.0mA					"	Q6	"		"		"		"
		38	"	"								"	"					-6.0mA	2.0V			"	Q7	"		"		"		"
		39	"	"								"	"							2.0V	-6.0mA	"	Q8	"		"		"		"
V_{OL6}	3007	40	"	20µA	0.8V							"	"									4.5V	Q1		0.1		0.1		0.1	"
		41	"	"		0.8V	20µA					"	"									"	Q2		"		"		"	"
		42	"	"				20µA	0.8V			"	"									"	Q3		"		"		"	"
		43	"	"						0.8V	20µA	"	"									"	Q4		"		"		"	"
		44	"	"								"	"	20µA	0.8V							"	Q5		"		"		"	"
		45	"	"								"	"			0.8V	20µA					"	Q6		"		"		"	"
		46	"	"								"	"					20µA	0.8V			"	Q7		"		"		"	"
		47	"	"								"	"							0.8V	20µA	"	Q8		"		"		"	"
V_{OL7}		48	"	6.0mA	0.8V							"	"									5.5V	Q1		0.26		0.4		0.26	"
		49	"	"		0.8V	6.0mA					"	"									"	Q2		"		"		"	"
		50	"	"				6.0mA	0.8V			"	"									"	Q3		"		"		"	"
		51	"	"						0.8V	6.0mA	"	"									"	Q4		"		"		"	"
		52	"	"								"	"	6.0mA	0.8V							"	Q5		"		"		"	"
		53	"	"								"	"			0.8V	6.0mA					"	Q6		"		"		"	"
		54	"	"								"	"					6.0mA	0.8V			"	Q7		"		"		"	"
		55	"	"								"	"							0.8V	6.0mA	"	Q8		"		"		"	"

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 1 T_C = +25°C		Subgroup 2 T_C = +125°C		Subgroup 3 T_C = -55°C		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
I_{OS4}	3011	56	GND	GND	4V	4V	GND	GND	4V	4V	GND	GND	3/	GND	4V	4V	GND	GND	4V	4V	GND	4V	Q1	-10	-135	-10	-135	-10	-135	mA
		57	"									"	"									"	Q2	"	"	"	"	"	"	"
		58	"									"	"									"	Q3	"	"	"	"	"	"	"
		59	"									"	"									"	Q4	"	"	"	"	"	"	"
		60	"									"	"									"	Q5	"	"	"	"	"	"	"
		61	"									"	"									"	Q6	"	"	"	"	"	"	"
		62	"									"	"									"	Q7	"	"	"	"	"	"	"
		63	"									"	"									"	Q8	"	"	"	"	"	"	"
I_{IH}	3010	64	5.5V		GND	GND			GND	GND			GND		GND	GND			GND	GND		5.5V	OE		.05		0.1			µA
		65	GND		5.5V	GND			"	"			"		"	"			"	"		"	D1		"		"			"
		66	"		GND	5.5V			"	"			"		"	"			"	"		"	D2		"		"			"
		67	"		"	GND			"	"			"		"	"			"	"		"	D3		"		"			"
		68	"		"	"			5.5V	5.5V			5.5V		"	"			"	"		"	D4		"		"			"
		69	"		"	"			GND	GND			GND		"	"			"	"		"	CLK		"		"			"
		70	"		"	"			"	"			"		5.5V	5.5V			"	"		"	D5		"		"			"
		71	"		"	"			"	"			"		GND	GND			"	"		"	D6		"		"			"
		72	"		"	"			"	"			"		"	GND			5.5V	5.5V		"	D7		"		"			"
		73	"		"	"			"	"			"		"	"			GND	"		"	D8		"		"			"
I_{IL}	3009	74	"		5.5V	5.5V			5.5V	5.5V			5.5V		5.5V	5.5V			5.5V	"		"	$\overline{OE}$		-.05		-0.1			"
		75	5.5V		GND	5.5V			"	"			"		"	"			"	"		"	D1		"		"			"
		76	"		5.5V	GND			"	"			"		"	"			"	"		"	D2		"		"			"
		77	"		"	5.5V			"	"			"		"	"			"	"		"	D3		"		"			"
		78	"		"	"			GND	GND			"		"	"			"	"		"	D4		"		"			"
		79	"		"	"			5.5V	5.5V			GND		"	"			"	"		"	CLK		"		"			"
		80	"		"	"			"	"			5.5V		GND	"			"	"		"	D5		"		"			"
		81	"		"	"			"	"			"		5.5V	GND			"	"		"	D6		"		"			"
		82	"		"	"			"	"			"		"	5.5V			GND	"		"	D7		"		"			"
		83	"		"	"			"	"			"		"	5.5V			5.5V	GND		"	D8		"		"			"
I_{OZH}		84	2.0V	2.0V	0.8V	0.8V	2.0V	2.0V	0.8V	0.8V	2.0V	"	3/	2.0V	0.8V	0.8V	2.0V	2.0V	0.8V	0.8V	2.0V	"	Q1		0.2		2.0			"
		85	"		"	"			"	"		"	"		"	"			"	"		"	Q2		"		"			"
		86	"		"	"			"	"		"	"		"	"			"	"		"	Q3		"		"			"
		87	"		"	"			"	"		"	"		"	"			"	"		"	Q4		"		"			"
		88	"		"	"			"	"		"	"		"	"			"	"		"	Q5		"		"			"
		89	"		"	"			"	"		"	"		"	"			"	"		"	Q6		"		"			"
		90	"		"	"			"	"		"	"		"	"			"	"		"	Q7		"		"			"
		91	"		"	"			"	"		"	"		"	"			"	"		"	Q8		"		"			"
I_{OZL}		92	"	GND	2.0V	2.0V	GND	GND	2.0V	2.0V	GND	"	"	GND	2.0V	2.0V	GND	GND	2.0V	2.0V	GND	"	Q1		-0.2		-2.0			"
		93	"		"	"			"	"		"	"		"	"			"	"		"	Q2		"		"			"
		94	"		"	"			"	"		"	"		"	"			"	"		"	Q3		"		"			"
		95	"		"	"			"	"		"	"		"	"			"	"		"	Q4		"		"			"
		96	"		"	"			"	"		"	"		"	"			"	"		"	Q5		"		"			"
		97	"		"	"			"	"		"	"		"	"			"	"		"	Q6		"		"			"
		98	"		"	"			"	"		"	"		"	"			"	"		"	Q7		"		"			"
		99	"		"	"			"	"		"	"		"	"			"	"		"	Q8		"		"			"
																								Subgroup 4 T_C = +25°C						
																								Min	Max					
C_C	3012	100	4/									GND										GND	$\overline{OE}$		15					pF
C_I		101			4/	4/						"										"	D1		10					"
		102										"										"	D2		"					"
		103							4/			"										"	D3		"					"
		104								4/		"										"	D4		"					"
C_C		105										"	4/									"	CLK		15					"
C_I		106										"			4/							"	D5		10					"
		107										"				4/						"	D6		"					"
		108										"							4/			"	D7		"					"
		109										"								4/			"	D8		"					"

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 4 T_C = +25°C					
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
C_O	3012	110	5.5V	4/			4/					GND										5.5V	Q1		20					pF
		111						4/				"										"	Q2		"					"
		112										"										"	Q3		"					"
		113									4/	"										"	Q4		"					"
		114										"		4/								"	Q5		"					"
		115										"					4/					"	Q6		"					"
		116										"						4/				"	Q7		"					"
		117										"									4/	"	Q8		"					"
																								Subgroup 7 T_C = +25°C		Subgroup 8 T_C = +125°C		Subgroup 8 T_C = -55°C		
																								Min	Max	Min	Max	Min	Max	
Truth table tests 5/ 6/	3014	118	B	X	A	A	X	X	A	A	X	GND	B	X	A	A	X	X	A	A	X	4.5V	All outputs 7/ 8/							
		119	"	H	B	B	H	H	B	B	H	"	A	H	B	B	H	H	B	B	H	"								
		120	"	"	"	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"	"									
		121	"	L	A	A	L	L	A	A	L	"	A	L	A	A	L	L	A	A	L	"								
		122	"	L	A	A	L	L	A	A	L	"	B	L	A	A	L	L	A	A	L	"								
		123	"	L	A	A	L	L	A	A	L	"	A	L	A	A	L	L	A	A	L	"								
		124	"	H	A	A	H	H	A	A	H	"	B	H	A	A	H	H	A	A	H	"								
																								Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
																								Min	Max	Min	Max	Min	Max	
f_{MAX} 9/		125	GND	OUT	IN							GND	IN									4.5V	Q1	28		21		28		MHz
		126	"			IN	OUT					"	"									"	Q2	"		"		"		"
		127	"					OUT	IN			"	"									"	Q3	"		"		"		"
		128	"							IN	OUT	"	"									"	Q4	"		"		"		"
		129	"									"	"	OUT	IN							"	Q5	"		"		"		"
		130	"									"	"			IN	OUT					"	Q6	"		"		"		"
		131	"									"	"					OUT	IN			"	Q7	"		"		"		"
		132	"									"	"							IN	OUT	"	Q8	"		"		"		"
t_{PHL2}	3003 (Fig. 3)	133	"	OUT	IN	4.5V	OUT		4.5V	4.5V		"	"		4.5V	4.5V			4.5V	4.5V		"	CLK to Q1		32		42		32	ns
		134	"		4.5V	IN			4.5V	"		"	"		"	"			"	"		"	CLK to Q2		"		"		"	"
		135	"		"	4.5V		OUT	IN	IN	OUT	"	"		"	"			"	"		"	CLK to Q3		"		"		"	"
		136	"		"	"			4.5V	"		"	"		IN	"			"	"		"	CLK to Q4		"		"		"	"
		137	"		"	"			"	4.5V		"	"	OUT	"	"			"	"		"	CLK to Q5		"		"		"	"
		138	"		"	"			"	"		"	"		4.5V	IN	OUT		"	"		"	CLK to Q6		"		"		"	"
		139	"		"	"			"	"		"	"		"	4.5V		OUT	IN	"		"	CLK to Q7		"		"		"	"
		140	"		"	"			"	"		"	"		"	4.5V			4.5V	IN	OUT	"	CLK to Q8		"		"		"	"
t_{PLH2}		141	"	OUT	IN	GND	OUT		GND	GND		"	"		GND	GND			GND	GND		"	CLK to Q1		"		"		"	"
		142	"		GND	IN			GND	"		"	"		"	"			"	"		"	CLK to Q2		"		"		"	"
		143	"		"	GND		OUT	IN	IN	OUT	"	"		"	"			"	"		"	CLK to Q3		"		"		"	"
		144	"		"	"			GND	"		"	"		"	"			"	"		"	CLK to Q4		"		"		"	"
		145	"		"	"			"	GND		"	"	OUT	GND	"			"	"		"	CLK to Q5		"		"		"	"
		146	"		"	"			"	"		"	"		"	IN	OUT		"	"		"	CLK to Q6		"		"		"	"
		147	"		"	"			"	"		"	"		"	GND		OUT	IN	"		"	CLK to Q7		"		"		"	"
		148	"		"	"			"	"		"	"		"	GND			GND	IN	OUT	"	CLK to Q8		"		"		"	"
t_{PZH}		149	IN	OUT	4.5V							"										"	$\overline{OE}$ to Q1		31		41		31	"
		150	"			4.5V	OUT					"										"	$\overline{OE}$ to Q2		"		"		"	"
		151	"					OUT	4.5V			"										"	$\overline{OE}$ to Q3		"		"		"	"
		152	"							4.5V	OUT	"										"	$\overline{OE}$ to Q4		"		"		"	"
		153	"									"		OUT	4.5V							"	$\overline{OE}$ to Q5		"		"		"	"
		154	"									"				4.5V	OUT					"	$\overline{OE}$ to Q6		"		"		"	"
		155	"									"						OUT	4.5V			"	$\overline{OE}$ to Q7		"		"		"	"
		156	"									"								4.5V	OUT	"	OE to Q8		"		"		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 52 – Continued.

Symbol	MIL-STD-883 method	Cases R,S, 2	Terminal conditions 1/																				Measured terminal	Test limits						Unit
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		Subgroup 9 T_C = +25°C 10/		Subgroup 10 T_C = +125°C		Subgroup 11 T_C = -55°C 10/		
		Test no.	$\overline{OE}$	Q1	D1	D2	Q2	Q3	D3	D4	Q4	GND	CLK	Q5	D5	D6	Q6	Q7	D7	D8	Q8	V_{CC}		Min	Max	Min	Max	Min	Max	
t_{PZL}	3003 (Fig. 3)	157	IN	OUT	GND	GND	OUT	OUT	GND	GND	OUT	GND		OUT	GND	GND	OUT	OUT	GND	GND	OUT	4.5V	$\overline{OE}$ to Q1		31		41		31	ns
		158	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		159	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		160	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		161	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		162	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		163	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		164	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{PHZ}		165	"	OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"		OUT	4.5V	4.5V	OUT	OUT	4.5V	4.5V	OUT	"	$\overline{OE}$ to Q1		"		"		"	"
		166	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		167	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		168	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		169	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		170	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		171	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		172	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{PLZ}		173	"	OUT	GND	GND	OUT	OUT	GND	GND	OUT	"		OUT	GND	GND	OUT	OUT	GND	GND	OUT	"	$\overline{OE}$ to Q1		"		"		"	"
		174	"									"										"	$\overline{OE}$ to Q2		"		"		"	"
		175	"									"										"	$\overline{OE}$ to Q3		"		"		"	"
		176	"									"										"	$\overline{OE}$ to Q4		"		"		"	"
		177	"									"										"	$\overline{OE}$ to Q5		"		"		"	"
		178	"									"										"	$\overline{OE}$ to Q6		"		"		"	"
		179	"									"										"	$\overline{OE}$ to Q7		"		"		"	"
		180	"									"										"	$\overline{OE}$ to Q8		"		"		"	"
t_{THL}	3004	181	GND	OUT	IN	4.5V	OUT		4.5V	4.5V		"	IN		4.5V	4.5V			4.5V	4.5V		"	Q1		12		16		12	"
		182	"		4.5V	IN			4.5V	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		183	"		"	4.5V		OUT	IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		184	"		"	"			4.5V	IN	OUT	"	"		"	"			"	"		"	Q4		"		"		"	"
		185	"		"	"			"	4.5V		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		186	"		"	"			"	"		"	"		4.5V	IN	OUT		"	"		"	Q6		"		"		"	"
		187	"		"	"			"	"		"	"		"	4.5V		OUT	IN	"		"	Q7		"		"		"	"
		188	"		"	"			"	"		"	"		"	4.5V			4.5V	IN	OUT	"	Q8		"		"		"	"
t_{TLH}		189	"	OUT	IN	GND	OUT		GND	GND		"	"		GND	GND			GND	GND		"	Q1		"		"		"	"
		190	"		GND	IN			GND	"		"	"		"	"			"	"		"	Q2		"		"		"	"
		191	"		"	GND		OUT	IN	"		"	"		"	"			"	"		"	Q3		"		"		"	"
		192	"		"	"			GND	IN	OUT	"	"		"	"			"	"		"	Q4		"		"		"	"
		193	"		"	"			"	GND		"	"	OUT	IN	"			"	"		"	Q5		"		"		"	"
		194	"		"	"			"	"		"	"		GND	IN	OUT		"	"		"	Q6		"		"		"	"
		195	"		"	"			"	"		"	"		"	GND		OUT	IN	"		"	Q7		"		"		"	"
		196	"		"	"			"	"		"	"		"	GND			GND	IN	OUT	"	Q8		"		"		"	"

1/ Output pins not designated may be "high" level logic, "low" level logic, or open. Input pins not designated may be "high" level logic or "low" level logic. Exceptions are as follows:

a. V_{IC}(pos) tests, the GND terminal shall be open.
b. V_{IC}(neg) tests, the V_{CC} terminal shall be open.
c. I_{CC} tests, the output terminal shall be open.

2/ For equipment that does not allow GND pin to be open during test, minimum limit of 0.4 V applies.

3/ Apply a positive going pulse. ⎍ 2.5 V minimum/5.5 V maximum.

4/ See 4.4.1c.

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5/ Tests performed in sequence.

6/ A = 3.7 V, B = 0.4 V for device types 01 through 06. A = 2.4 V, B = 0.4 V for device type 52.

7/ X = Indeterminate output voltage.

8/ Output shall be: High $\geq$ 1.5 V; Low $\leq$ 1.5 V, $\left| I_{OUT} \right|$ = 20 μA.

9/ f_{MAX}, minimum limit specified, is the frequency of the input pulse. The output frequency shall be one-half of the input frequency. The f_{MAX} requirement is considered met if proper output state changes occur with the pulse repetition period set to that given in the limits column.

10/ See 4.4.1d.

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4.5 Methods of inspection. Methods of inspection shall be specified and as follows:

4.5.1 Voltage and current. Unless otherwise specified, all voltages given are referenced to the microcircuit GND terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When the burn-in and life tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to within 10°C of their power stable condition at room temperature; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits at 25°C.

Parameter 1/	Device types	
	All	
I_{CC}		±30 nA

1/ The above parameters shall be recorded before and after the required burn-in and life tests to determine deltas (Δ).

4.5.3 Quiescent supply current (I_{CC} test). When performing quiescent supply current measurements (I_{CC}), the meter shall be placed so that all currents flow through the meter.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. PIN and compliance identifier, if applicable (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractors parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535, MIL-HDBK-1331, and as follows:

C_{IN}..... Input terminal-to-GND capacitance.
GND ..... Ground zero voltage potential
I_{CC} ..... Quiescent supply current.
T_A ..... Free air temperature.
V_{CC} ..... Positive supply voltage.

6.6 Logistic support. Lead materials and finishes (see 3.4) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

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6.7 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, will be supplied.

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady-state life tests (see 3.6).

b. A copy of each radiograph.

c. The technology conformance inspection (TCI) data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parameters data (see 4.2d).

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

Military device type	Generic-industry type
01	54HC273
02	54HC374
03	54HC377
04	54HC574
05	54HC534
06	54HC564
52	54HCT374

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5962-2005-022)

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at. http://assist.daps.dla.mil

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